

STAFFORD’S CONJECTURE ON CYCLICITY OF TORSION MODULES

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ABSTRACT. Let $A_n(k)$ be the Weyl algebra over a field of characteristic zero. We prove that every nonzero $d \in A_n(k)$ admits $F, R, S \in A_n(k)$ such that $1 = dR + FdS$. Thus Stafford’s Conjecture 3.8 holds. For $n \geq 1$, the multiplier F may be chosen as x^N , after a linear symplectic change of variables, where N is the Bernstein degree of d . The proof studies a cyclic quotient through its characteristic variety and excludes nonzero quotients by conormal geometry at infinity.

AI comment Review guide. **Johanna:** review the red text and proposed corrections as mathematics and prose; no Lean reading is required. **Max:** review the whole selected paper proof against its checked Lean route, including definitions and exact matches. The interface and paper use the same frozen revisions. Readable correspondence notes are optional aids, not another review assignment. Red text awaits author acceptance; blue strike-through preserves small original passages. Replaced long proofs remain verbatim in marked source-history comments and Git history. Historical alternatives are outside the selected publication proof. Resolve a box after its task is done.

AI comment AUDIT-01. Information — no edit requested. Corrections, clarifications and mathematical differences from the formal proof remain marked as AI additions. Long formal proof comparisons and declaration links are collected in the separate Overleaf supplement `lean_proof_details.tex`. The human-readable draft shows the mathematical qualifications beside the relevant passages; the side-by-side audit retains the full technical correspondence.

1. INTRODUCTION

Let k be a field of characteristic zero, and let

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n \rangle$$

be the n th Weyl algebra, with $[\partial_i, x_j] = \delta_{ij}$ and $[\partial_i, \partial_j] = [x_i, x_j] = 0$.

Stafford conjectured in 1978 (Conjecture 3.8 of [30, p. 438]) that for every nonzero $d \in A_n(k)$, one can find $F \in A_n(k)$ with $A_n(k) = dA_n(k) + FdA_n(k)$. Stafford notes that this immediately implies $A_n(k)/dA_n(k)$ is cyclic, i.e. generated by a single element. [7] reported the resulting question—whether every right torsion module over $A_n(k)$ is cyclic—as open [Section 3, Conjecture 3.4]. We prove Stafford’s conjecture:

Stafford’s Conjecture 3.8 asks whether every nonzero $d \in A_n(k)$ admits $F \in A_n(k)$ with $A_n(k) = dA_n(k) + FdA_n(k)$ [30, p. 438]. Stafford identifies

this with stable cyclicity for finitely generated torsion right Weyl-algebra modules [30, pp. 437–438]. Bellamy’s survey states the weaker ordinary-cyclicity question for finitely generated torsion $A_n(\mathbb{C})$ -modules as open [7, Section 3, Conjecture 3.4]. We prove Stafford’s conjecture:

AI comment INTRO-01. Proposed change — author acceptance pending.

The quotient A_n/dA_n is always cyclic, generated by the class of 1. The nontrivial implication concerns general *finitely generated* torsion modules. Bellamy states the ordinary cyclicity question over $\mathbb{C}$; Stafford’s stable-cyclicity formulation is stronger. The red replacement above restores these distinctions.

Theorem 1.1. *For every characteristic-zero field k , every $n \geq 1$, and every nonzero $d \in A_n(k)$, there exist $F, R, S \in A_n(k)$ such that*

$$1 = dR + FdS.$$

Moreover, one may choose $F = \ell^{\deg_B d}$ for a linear Weyl coordinate ℓ , obtained by an invertible linear symplectic change of generators. Explicitly, $\ell = \Phi_g^{-1}(x_1)$ for a linear ω -preserving automorphism g as in Section 4.

Formal comparison: Lean also covers $n = 0$, where $A_0(k) = k$

The machine-checked proof of this result is registered in Palomar [5, 6]. Its Global Stafford extension [3, 4] proves the same-divisor identity $1 = dR + FdS$ in $D_k(A)$ for every smooth integral finitely generated characteristic-zero k -algebra A and every nonzero finite-order k -linear differential operator d . Polynomial A recovers the Weyl-algebra identity; the global theorem asserts existence of F , without the fixed linear-coordinate refinement in Theorem 1.1.

Proof sketch: Give each standard generator Bernstein degree one and let $N = \deg_B d$. A linear symplectic change of variables makes d monic of degree N in a momentum p , whose conjugate coordinate we denote by x . It is enough to prove that the cyclic right module

$$(1.1) \quad Q = A_n/(dA_n + x^N dA_n).$$

vanishes. An Euler identity makes x surjective on Q , while monicity makes $H = \{x = 0\}$ noncharacteristic and places $\text{Char}(Q)$ in a hypersurface $V(P)$ with $P(dx) \neq 0$. A filtered-page length argument shows that $\text{Char}(Q)$ has no point over H . Gabber’s theorem and the conormal argument of Sections 4 and 5 then force $\text{Char}(Q)$ to be empty. The same argument gives the more precise identity $A_n = dA_n + x^N dA_n$ in the chosen symplectic coordinates.

Constructive predecessors concern Stafford’s established generator theorems. Hillebrand–Schmale and Leykin give effective two-generator methods, and Leykin also constructs cyclic generators of holonomic modules [17, 20]. Quadrat–Robertz develop module-decomposition and basis algorithms [26, 27]; Caro–Levcovitz and Caro–Tuesta–Levcovitz study differential-operator rings with series coefficients [9, 10]. The constrained identity for arbitrary nonzero d requires the additional quotient-vanishing argument developed here.

The surrounding generator theory includes Stafford's stable-range and noncommutative Forster–Swan results [32, 31]. The generator article has a corrigendum and related exposition [34, 33]. Stafford also constructed stably free Weyl-algebra ideals [35]. OreModules supplies computational methods for modules over Ore algebras [11]. These provide context; the Euler residue, filtered-length contradiction, and asymptotic conormal construction below are project arguments.

2. ~~AI STATEMENT~~ [CHRIS] FORMAL VERIFICATION

~~Every statement of this paper, including Theorems 8.2 and 9.2 at the generality stated, is formalized and verified in Lean 4 against Mathlib, with no axiom beyond Lean's three standard ones.~~ The main identity, its fixed-source strengthening and the documented cyclicity and noncharacteristic consequences are verified in Lean 4 against Mathlib using only the three standard axioms. The formal proof also supplies the characteristic-support and geometric results used by its own argument. Statement scope and proof routes must be compared separately: machine verification of a terminal theorem does not certify every intermediate step of the printed proof. The mathematical differences are noted below; full formal comparisons and declaration links are in the technical supplement. The headline theorem is registered in Palomar [5]; the current Lean sources and auxiliary results are archived [6]. The accompanying proof-map supplement names the Lean declaration for each step.

AI comment WF-01. Proposed change — author acceptance pending. Please review each comparison at the stated generality. The Palomar record checks the registered main theorem; later results have separate archived sources. The formal geometric route uses Laurent series, while the printed tangent-limit argument is auxiliary. The short comparison notes identify these differences without claiming that every numbered statement or every proof passage has been verified verbatim.

The development uses Lean 4 [14], Mathlib [36] and AlgebraicAnalysis v0.3.0 [1]; tooling, chronology and statement comparison are described in Appendix A.

Online resources. Online Resource 1 is the interactive paper–Lean correspondence review, including the exact challenge definitions and archived proof endpoints; Online Resource 2 is the readable mathematical account of the selected formal proofs. Both are available from the [review companion](#), with a [frozen supplementary package](#) for journal review. The reusable HTML generator is maintained separately under the Apache 2.0 license; its software archive and the manuscript supplement have distinct version records. Proposed corrections and unresolved review actions remain explicitly marked in this draft.

3. NOTATION AND DEFINITIONS

Throughout, k is a field of characteristic zero and for the n -th Weyl algebra over k , we write

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n \rangle, \quad [\partial_i, x_j] = \delta_{ij}, \quad [x_i, x_j] = [\partial_i, \partial_j] = 0.$$

All ideals are *right* ideals and all modules are *right* modules unless the contrary is said explicitly.

Convention 3.1. *We single out the first Weyl pair (x_1, ∂_1) and write*

$$p := \partial_1, \quad C := A_{n-1}(k)[x_1], \quad A_n(k) = C[p; \underline{\partial_1}\delta],$$

Here $\delta(x_1) = 1$ and δ vanishes on $A_{n-1}(k)$. so that x_1 is central in C , $[p, x_1] = 1$, and the transverse algebra $A_{n-1}(k)$ commutes with x_1 and p . Every element of $A_n(k)$ can uniquely be written as $\sum_j \underline{p^j} c_j p^j$ with $c_j \in C$.

This is the derivation Ore presentation; Ore[24] supplies its historical algebraic framework.

Formal comparison: coefficients on the left, $\sum_j c_j p^j$; both normal forms are unique

Convention 3.2. *We define $H := \{x_1 = 0\} = \{(0, x_2, \dots, x_n)\} \subseteq \mathbb{A}^n$ as a distinguished hyperplane; its conormal covector is dx_1 , i.e. $\xi = (1, 0, \dots, 0)$. We write $\pi: T^*\mathbb{A}^n \rightarrow \mathbb{A}^n$ for the projection and identify ξ_i with dx_i , so that a covector is $\sum_i \xi_i dx_i$. The hyperplane $H = \{x_1 = 0\}$, the projection π and the identification $\xi_i \leftrightarrow dx_i$ are fixed in Section 3.1 below.*

AI comment DUP-01. Proposed change — author acceptance pending. Convention 3.2 repeated the definitions of H , π and ξ_i given again in Section 3.1; the red text keeps one copy. Alternatively delete the whole convention environment.

3.1. Cotangent space and symplectic structure. Let $X = \mathbb{A}_k^n$ be affine n -space over a field k of characteristic zero, with coordinates $x_1, \dots, x_n$. Its cotangent bundle T^*X is the space of pairs (y, ξ) consisting of a point $y \in X$ together with a covector ξ at y ; since X is affine space, this cotangent space is the same vector space $(k^n)^*$ at every point, and we may identify

$$T^*X \cong X \times (k^n)^*.$$

We write $\xi_1, \dots, \xi_n$ for the coordinates on $(k^n)^*$ dual to $x_1, \dots, x_n$, so that a covector is $\sum_i \xi_i dx_i$, and identify $\text{gr}^B A_n(k) = k[x, \xi]$ with the coordinate ring of T^*X . We write $\pi: T^*X \rightarrow X$ for the projection forgetting the covector, and for $y \in X$ write $T_y^*X = \pi^{-1}(y) \cong (k^n)^*$ for the fibre of T^*X over y .

We single out the first coordinate x_1 and define the distinguished hyperplane

$$H := \{x_1 = 0\} = \{(0, x_2, \dots, x_n)\} \subseteq X;$$

its conormal covector is dx_1 , i.e. $\xi = (1, 0, \dots, 0)$.

Definition 3.3 (Filtration [Def. 2.6 and 2.9 [28]]). A filtration $F_\bullet = A_n(k)F_\bullet F_\bullet$ on $A_n(k)$ is a sequence of linear subspaces

$$\{0\} = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq \dots \subseteq A_n(k),$$

such that $F_i \cdot F_j \subseteq F_{i+j}$ and $A_n(k) = \bigcup F_j$. The associated graded algebra, also called symbol ring, is defined as

$$\mathrm{gr}^F A_n(k) = \bigoplus_{j=0}^{\infty} F_j / F_{j-1}.$$

Let $a \neq 0 \in A_n(k)$ and let j be the smallest integer, such that $a \in F_j$, called a filtration degree $\deg_F a$. ~~The~~ ~~the~~ The class of a in F_j / F_{j-1} is called its symbol $\sigma_j(a)$.

In the proof, we use two filtrations of $A_n(k)$.

- The *Bernstein filtration* $B_\bullet$ assigns degree 1 to every x_i and every ∂_i . The Bernstein degree of d is written $\deg_B d$.
- The *order filtration* $F_\bullet$ assigns degree 0 to every x_i and degree 1 to every ∂_i ; F_m is spanned by the monomials $x^\alpha \partial^\beta$ with $|\beta| \leq m$.

Formal comparison: as spans of ordered PBW words of bounded weight

For both the order filtration and the Bernstein filtration, the associated graded algebra $\mathrm{gr} A_n(k)$ is simply the polynomial ring $k[x, \xi] = k[x_1, \dots, x_n, \xi_1, \dots, \xi_n]$ in $2n$ variables with ξ_i the symbol of ∂_i . In particular, $\mathrm{gr} A_n(k)$ is commutative [28, Prop. 2.10]. Good filtrations and characteristic varieties are treated systematically in Björk and Hotta–Takeuchi–Tanisaki[8, 18].

Definition 3.4. A filtration of a right A_n -module M compatible with $F_\bullet$ is an exhaustive increasing chain of $k_0 k$ -subspaces $M_\bullet$ with $M_i F_j \subseteq M_{i+j}$ and $M_i = 0$ for $i < 0$. It is good if the associated graded module $\mathrm{gr} M = \bigoplus_i M_i / M_{i-1}$ is finitely generated over $\mathrm{gr}^F A_n$.

As an example, the filtration $M_i := qF_i$ of a cyclic module over A_n induced by its generator q is good, since $\mathrm{gr} Q$ is generated by the element $\bar{q}$ denoting the class of q in Q_0 / Q_{-1} .

For a finitely generated right A_n -module M with a good order filtration $M_\bullet$, the graded module $\mathrm{gr} M$ is a finitely generated $k_0[x, \xi] k[x, \xi]$ -module.

Definition 3.5. The characteristic variety (also called symbol variety) of a finitely generated A_n -module M with a good order filtration $M_\bullet$ is defined by

$$\mathrm{Char}(M) := V(\mathrm{Ann}_{k[x, \xi]} \mathrm{gr} M) \subseteq T^* \mathbb{A}^n = \mathrm{Spec} k[x, \xi],$$

$\mathrm{Ann}_{k[x, \xi]} \mathrm{gr} M = \{f \in k[x, \xi] : f \cdot m = 0 \forall m \in \mathrm{gr} M\}$ are the annihilators of $\mathrm{gr} M$ over $k[x, \xi]$, and V denotes their vanishing locus, i.e. the set of points (x, ξ) for which all annihilator polynomials vanish. This set of points lies in the cotangent bundle $T^* \mathbb{A}^n$ of the affine n -space $\mathbb{A}^n$.

~~$\mathrm{Char}(M)$ is independent of the choice of the good filtration.~~ We only use $M = A_n / I$ with the filtration $M_i = qF_i$ induced by the class q of 1; then $\mathrm{Ann} \mathrm{gr} M$ is the order initial ideal of I , spanned by the order symbols of the

elements of I . Points are prime ideals; for a non-closed field this is the correct reading of “the set of points”.

Formal comparison: for $M = A_n/I$

Definition 3.6 (Symplectic form and Poisson bracket). *A bilinear form ω on a k -vector space is symplectic if it is antisymmetric and nondegenerate.*

The standard symplectic form on the vector space k^{2n} , with basis $x_1, \dots, x_n, \partial_1, \dots, \partial_n$, is determined by the relations

$$\omega(x_i, \partial_j) = \delta_{ij}, \quad \omega(x_i, x_j) = \omega(\partial_i, \partial_j) = 0,$$

*analogously to the Weyl relations. Identifying $T^*X \cong X \times (\mathbb{C}k^n)^*$ via the coordinates $x_1, \dots, x_n$ and their duals $\xi_1, \dots, \xi_n$, on T^*X this ω takes the familiar form*

$$\omega = \sum_{i=1}^n dx_i \wedge d\xi_i.$$

*The associated Poisson bracket on $k[x, \xi]$, the coordinate ring of T^*X , is*

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial \xi_i} \frac{\partial g}{\partial x_i} - \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial \xi_i} \right).$$

Formal comparison: Lean’s bracket is the negative of this one; involutivity is unaffected

Definition 3.7 (involutive). *A reduced closed subset of $T^*\mathbb{A}^n$ is involutive if its vanishing ideal is closed under the Poisson bracket, $\{I(W), I(W)\} \subseteq I(W)$. An ideal J is involutive if $\{J, J\} \subseteq J$. This is weaker than being a Poisson ideal ($\{J, k[x, \xi]\} \subseteq J$): the ideal $(\xi_1, \dots, \xi_n)$ is involutive but not Poisson.*

AI comment ALG-01. Proposed change — author acceptance pending. Gabber applies to the radical characteristic ideal: put $J = \text{Ann}_{k[x, \xi]} \text{gr } M$ and $I = \sqrt{J}$; then $\{I, I\} \subseteq I$. The set $\text{Char}(M) = V(J)$ is not an ideal. Use $I = \sqrt{J}$ in the componentwise lemma and its proof below. Minimal primes of J and $\sqrt{J}$ coincide, so the localization argument then has the intended hypotheses.

3.2. Gabber’s theorem.

Theorem 3.8 (Gabber [15]; algebraic form in SinghKumar[29]; see also [18, §2.4 Theorem 2.3.1 and Appendix D]–Björk). *Let I be any right ideal of $A_n(k)$, with k a field of characteristic zero, and give $M = A_n(k)/I$ its order quotient filtration. Then, the characteristic variety $\text{Char}(M)$ is an involutive ideal, i.e. $\{\text{Char}(M), \text{Char}(M)\} \subseteq \text{Char}(M)$. the radical $\sqrt{J}$ of $J = \text{Ann}_{k[x, \xi]} \text{gr } M$ is involutive, $\{\sqrt{J}, \sqrt{J}\} \subseteq \sqrt{J}$; equivalently, $\text{Char}(M) = V(J)$ is involutive in the sense of Definition 3.7.*

Formal comparison: proved in Lean, not cited; exactly this scope, over every field of characteristic zero

AI comment LIT-01. Information — no edit requested. Statement check: Gabber [15] proves that $\sqrt{\text{Ann gr } M}$ is closed under the Poisson bracket for every finitely generated module M with a good filtration over a filtered ring whose associated graded ring is commutative, Noetherian and contains $\mathbb{Q}$; Singh–Kumar give the algebraic proof. The red statement above is the special case $M = A_n(k)/I$ with the quotient filtration, which is all the paper uses and exactly what the formalization proves (following the Singh–Kumar argument: Rees ring modulo T^2 , localization at a minimal prime, Artinian truncation, trace identity).

Lemma 3.9 (Componentwise involutivity). *Let M be a finitely generated $A_n(k)$ -module with a good filtration, and let $\text{Char}(M)$ be its characteristic variety, so that by Gabber's theorem $\{\text{Char}(M), \text{Char}(M)\} \subseteq \text{Char}(M)$. If $\mathfrak{p}$ is a minimal prime of $\text{Ann}_{k[x, \xi]} \text{gr } M$ (equivalently, a minimal prime over $\text{Char}(M)$), let J be an ideal of $k[x, \xi]$ whose radical is involutive (by Theorem 3.8, e.g. $J = \text{Ann gr } M$ for $M = A_n(k)/I$). If $\mathfrak{p}$ is a minimal prime of J , then $\mathfrak{p}$ is itself involutive:*

$$\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}.$$

AI addition: Suggested proof matching the checked route.

Proof. Lean proves the lemma in the corrected form of ALG-01. Let J be any ideal of $R = k[x, \xi]$ with $\{\sqrt{J}, \sqrt{J}\} \subseteq \sqrt{J}$, and let $\mathfrak{p}$ be minimal over J . Then $\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}$. The proof avoids localization.

Saturation. For $f \in \mathfrak{p}$ there is $s \notin \mathfrak{p}$ with $sf \in \sqrt{J}$. Otherwise, let S be the multiplicative set of all products sf^m with $s \notin \mathfrak{p}$ and $m \geq 0$. Then S is disjoint from J . For $m = 0$ this is because $J \subseteq \mathfrak{p}$. For $m \geq 1$, if $sf^m \in J$ then $(sf)^m = s^{m-1} \cdot sf^m \in J$, so $sf \in \sqrt{J}$, contrary to the assumption. Hence some prime $\mathfrak{Q} \supseteq J$ is disjoint from S . It satisfies $\mathfrak{Q} \subseteq \mathfrak{p}$, so $\mathfrak{Q} = \mathfrak{p}$ by minimality. But $f \in \mathfrak{p} \cap S$, a contradiction.

Leibniz step. Let $f, g \in \mathfrak{p}$. Choose $s, t \notin \mathfrak{p}$ with $sf, tg \in \sqrt{J}$, and put $u = st \notin \mathfrak{p}$. Then $uf, ug \in \sqrt{J}$. Since $\{\cdot, \cdot\}$ is a biderivation and $\{u, u\} = 0$,

$$u^2\{f, g\} = \{uf, ug\} - ug\{f, u\} - fu\{u, g\}.$$

The first term lies in $\sqrt{J} \subseteq \mathfrak{p}$ by involutivity, and the other two lie in $\mathfrak{p}$ because $f, g \in \mathfrak{p}$. As $u^2 \notin \mathfrak{p}$, we get $\{f, g\} \in \mathfrak{p}$. $\square$

Formal comparison: any ideal with involutive radical; proof without localization

AI comment COMP-01. Proposed change — review the red proof. The shorter saturation and Leibniz proof now follows the checked argument. Radical involutivity is the hypothesis; no localization of the bracket is needed. [Readable correspondence note.](#)

4. MAPPING TO A MONIC NORMAL FORM

Given a fixed nonzero $d \in A_n(k)$ of Bernstein degree N , our first goal is to find a map to replace d by a normal form $\tilde{d}$ that is monic in p , i.e. of the form $\tilde{d} = p^N + \sum_{j < N} p^j a_j$ with coefficients a_j in C . ~~$\tilde{d} = p^N + \sum_{j < N} p^j a_j$ with coefficients a_j in C~~ $\tilde{d} = p^N + \sum_{j < N} c_j p^j$ with coefficients c_j in C .

If $N = 0$, then $d \in B_0 = k$ is a nonzero scalar, hence a unit, and $1 = dR + FdS$ holds trivially with $R = d^{-1}, F = 0, F = 1 = \ell^0, R = d^{-1}, S = 0$. We therefore assume $N \geq 1$.

AI comment ALG-07. Proposed change — author acceptance pending.
 In the scalar case choose $F = 1, R = d^{-1}, S = 0$ to satisfy the stronger assertion $F = \ell^0 = 1$ too.

We want to construct a well-behaved automorphism of $\text{gr}^B A_n(k) = k[x, \xi]$ from simpler linear algebra. We proceed in three stages: define an automorphism of the vector space $\mathbb{V} = k^{2n}$, extend it to an automorphism of $A_n(k)$ itself, and observe the induced action this extension leaves on the associated graded ring.

We can identify $\mathbb{V} = k^{2n}$ with the Bernstein-degree-one part B_1/B_0 of $A_n(k)$, i.e. linear combinations of $x_1, \dots, x_n, \partial_1, \dots, \partial_n$; the coordinates of a vector are then the coefficients of these generators. By this identification, $\mathbb{V}$ carries the standard symplectic form ω (Def. 3.6). Let g be a k -linear automorphism of $\mathbb{V}$ that preserves the symplectic form, that is $\omega(gu, gv) = \omega(u, v)$ for all $u, v \in \mathbb{V}$.

Every such g extends uniquely to a k -algebra automorphism Φ_g of $A_n(k)$: substituting each generator $u \in \mathbb{V}$ by its image $g(u)$ keeps the Weyl relations $[u, v] = \omega(u, v)[v, u] = \omega(u, v)$ (recall $[\partial_i, x_i] = 1 = \omega(x_i, \partial_i)$) intact exactly when g preserves ω , and this substitution on generators determines Φ_g on all of $A_n(k)$.

Finally, since Φ_g sends each generator to another degree-one element, it preserves the Bernstein filtration, and therefore induces a well-defined automorphism of the associated graded ring $\text{gr}^B A_n(k) = k[x, \xi]$. ~~Concretely, this induced map sends a symbol (a polynomial function on $\mathbb{V}$) s to $s \circ g^{-1}$: functions transform by precomposition with the inverse, since Φ_g moves the underlying points of $\mathbb{V}$ forward by g .~~ Concretely, write the generators as $z_1, \dots, z_{2n}$ and let G be the matrix of g , $g(z_i) = \sum_j G_{ji} z_j$. A symbol is a polynomial $s(z)$, and Φ_g substitutes $z_i \mapsto \sum_j G_{ji} z_j$. Reading s as a function of coefficient vectors $c \in k^{2n}$, the symbol of $\Phi_g(d)$ is therefore $c \mapsto s(G^T c)$, i.e. $s \circ g^T$ (not $s \circ g^{-1}$). For homogeneous s of degree N , the value at the basis vector e_p is the coefficient of z_p^N .

AI comment ALG-08. Proposed change — author acceptance pending.

The printed law $s \mapsto s \circ g^{-1}$ is the point convention $\Phi(z) = z \circ g^{-1}$, not the substitution $\Phi_g(u) = g(u)$ defined above. Counterexample to the printed proof of Proposition 4.2: $n = 1$, $d = x$, $v = x + \partial$ (so $s(v) = 1$), $g = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, which preserves ω and maps v to $e_p = \partial$; yet $\Phi_g(x) = x$ has no p -term. The red replacement and the corrected proof below use $s \circ g^T$, as the formalization does.

Lemma 4.1. *Let $(\mathbb{V}, \omega)$ be a symplectic k -space of dimension $2n$ and $0 \neq v \in \mathbb{V}$. Let $\mathbb{V} = k^{2n}$ carry the standard symplectic form fixed above, and let $0 \neq v \in \mathbb{V}$. Then v is the first vector of a symplectic basis of $\mathbb{V}$; equivalently, there exists a k -linear g preserving ω with $g(v) = e$ for any prescribed nonzero e . For any prescribed nonzero e , there exists a k -linear automorphism g preserving ω with $g(v) = e$.*

AI addition: Suggested proof matching the checked route.

Proof. Use the pairing $\langle u, w \rangle = \sum_i (u_{\xi_i} w_{x_i} - u_{x_i} w_{\xi_i})$, which is $-\omega$ and has the same isometry group. For $a \in k$ and $u \in \mathbb{V}$ put $\tau_{a,u}(z) := z + a\langle z, u \rangle u$. Expanding, $\langle \tau_{a,u} z, \tau_{a,u} w \rangle = \langle z, w \rangle + a\langle z, u \rangle \langle u, w \rangle + a\langle w, u \rangle \langle z, u \rangle = \langle z, w \rangle$, since $\langle u, u \rangle = 0$ and $\langle u, w \rangle = -\langle w, u \rangle$; so every $\tau_{a,u}$ preserves the form. Its inverse is $\tau_{-a,u}$, hence it is a symplectic automorphism. Let $e, v \neq 0$.

Case $\langle e, v \rangle \neq 0$. Put $\tau_{e \rightarrow v} := \tau_{a, v-e}$ with $a = \langle e, v \rangle^{-1}$. Since $\langle e, v-e \rangle = \langle e, v \rangle$, we get $\tau_{e \rightarrow v}(e) = e + (v-e) = v$.

Case $\langle e, v \rangle = 0$. Choose a bridge vector w with $\langle e, w \rangle \neq 0 \neq \langle w, v \rangle$. Nondegeneracy (a suitable basis vector suffices) gives r, s with $\langle e, r \rangle \neq 0$ and $\langle s, v \rangle \neq 0$. Take $w = r$ if $\langle r, v \rangle \neq 0$, and $w = s$ if $\langle e, s \rangle \neq 0$. Otherwise take $w = r + s$, for which $\langle e, r+s \rangle = \langle e, r \rangle$ and $\langle r+s, v \rangle = \langle s, v \rangle$. Then the product of the two transvections $\tau_{w \rightarrow v} \tau_{e \rightarrow w}$ sends e to v . $\square$

Formal comparison: second formulation only; the formal proof composes at most two symplectic transvections

AI comment COMP-02. Proposed change — review the red statement and proof. The needed standard-space transitivity is proved by at most two explicit symplectic transvections. This avoids the unused general symplectic-basis construction. **Readable correspondence note.**

Proposition 4.2. *Let $0 \neq d \in A_n(k)$ with $N := \deg_B d \geq 1$. There exists a k -linear ω -preserving automorphism g and $c \in k^\times$ such that*

$$(4.1) \quad \tilde{d} := c^{-1} \Phi_g(d) = p^N + \sum_{j < N} c_j p^j, \quad c_j \in C, \quad \deg_B c_j \leq N - j,$$

in the notation of Convention 3.1.

AI addition: Suggested proof matching the checked route.

Proof. Let $s \in k[x, \xi]$ be the Bernstein principal symbol of d , a nonzero homogeneous polynomial of degree N on $\mathbb{V}$. A characteristic-zero field is infinite, so every nonzero polynomial has a nonvanishing value: choose $v \in \mathbb{V}$

with $s(v) \neq 0$. By Lemma 4.1 there is an ω -preserving h with $h(e_1) = v$, where e_1 is the basis vector in the p -direction. Let g be the map with matrix $H^\top$; the transpose of a symplectic matrix is symplectic, so g preserves ω . The Bernstein symbol of $\Phi_g(d)$ is $s \circ g^\top = s \circ h$, whose value at e_1 is $s(v) \neq 0$. Put $c = s(v)$; since c is a nonzero scalar, it is a central unit of $A_n(k)$, and $\tilde{d} = c^{-1}\Phi_g(d)$ has its Bernstein symbol taking the value 1 at e_1 . Since e_1 has coordinate 1 in the p -direction and 0 elsewhere, that value is precisely the coefficient of the monomial p^N in $\tilde{d}$. Collecting the remaining terms by their p -degree and writing the coefficients on the left, using the unique normal form $A_n(k) = \bigoplus_j Cp^j$, gives $\tilde{d} = p^N + \sum_{j < N} c_j p^j$ with $c_j \in C$. The Bernstein bound $\deg_B c_j \leq N - j$ holds because $\deg_B \tilde{d} = N$ and Φ_g preserves $\deg_B$; no cancellation between different p -degrees occurs, because the normal form $\bigoplus_j Cp^j$ is unique. $\square$

Formal comparison: monicity is stated as: $\tilde{d} \in B_N$ and the PBW coefficient of ∂_1^N is 1

AI comment COMP-03. Information — conventions aligned. The normal form now puts coefficients on the left, as in the checked PBW representation. Symbol substitution remains $s \circ g^\top$; the Bernstein bound gives the displayed coefficient bounds. [Readable correspondence note.](#)

All coefficients involved belong to a finitely generated subfield $k_0 \subset k$, namely the subfield generated by the finitely many coefficients of d in the new generators. If we find $R, S, F \in A_n(k_0)$ sufficing Theorem 1.1, the same choices extend to k as well, so proving the identity over k_0 is sufficient. We therefore replace k by k_0 , write $A_n := A_n(k_0)$, and fix an embedding $k_0 \hookrightarrow \mathbb{C}$. Sections 4–7 work over k itself; to keep the printed formulas we write $k_0 := k$ and $A_n := A_n(k_0)$ there. Sections 8 and 9 work over an algebraically closed field K of characteristic zero; the reader may take $K = \mathbb{C}$, and Section 10 uses $K = \bar{k}$. No embedding of k into $\mathbb{C}$ is needed.

AI comment FIELD-01. Proposed change — author acceptance pending. The formalization never reduces to a finitely generated subfield: it runs Sections 8–9 over k and descends the order initial ideal (Section 10). The red text adopts this route; the struck text is no longer needed.

Remark 4.3. Φ_g is a ring automorphism, so it preserves written order term by term. Consequently an identity $1 = \tilde{d}R + x_1^N \tilde{d}S$ transports back to

$$1 = d(c^{-1}\Phi_g^{-1}(R)) + F d(c^{-1}\Phi_g^{-1}(S)), \quad F = \Phi_g^{-1}(x_1)^N,$$

using that c^{-1} is a central scalar.

We now use the order filtration on $A_n(k_0)$, where coordinates have degree zero and derivatives have degree one, to characterize $P = \sigma_{\text{ord}}(\tilde{d})$, the order principal symbol of $\tilde{d}$. Recall that a symbol, being the class of an element in a single graded piece, is automatically *homogeneous* of the corresponding degree, i.e. contains only terms of that total degree.

Definition 4.4. We say a polynomial $P = \sum_{i=0}^N c_i \xi_1^i \in k_0[\xi_1, \dots, \xi_n]$, written as a polynomial in ξ_1 with coefficients $c_i \in k_0[\xi_2, \dots, \xi_n]$, is monic of degree N in ξ_1 if $c_N = 1$.

Lemma 4.5. Let $\tilde{d} = p^N + \sum_{j < N} c_j p^j$ with $c_j \in C$, $\deg_B c_j \leq N - j$, as in (4.1), and set $N = \deg_B \tilde{d}$. Then $\tilde{d}$ has order N , and its order principal symbol is a polynomial in ξ alone,

$$(4.2) \quad \sigma_{\text{ord}}(\tilde{d}) = P(\xi), \quad P \in k_0[\xi_1, \dots, \xi_n], \quad \deg P = N.$$

P is monic of degree N in ξ_1 and in particular

$$(4.3) \quad P(1, 0, \dots, 0) = 1.$$

Proof. Since $\deg_B \tilde{d} = N$, every PBW monomial $x^\alpha \partial^\beta$ occurring in $\tilde{d}$ satisfies $|\alpha| + |\beta| \leq N$. A monomial contributing to the order- N part has $|\beta| = N$, hence $|\alpha| = 0$: it is a pure ∂^β with scalar coefficient. Thus $\tilde{d}$ has order N , and its order- N part is a k_0 -linear combination of ∂^β with $|\beta| = N$, giving $\sigma_{\text{ord}}(\tilde{d}) = P(\xi)$, homogeneous of degree N with coefficients in k_0 .

The monomial $p^N = \partial_1^N$ occurs in $\tilde{d}$ with coefficient 1; every other order- N monomial comes from some $p^j a_j$ with $j < N$ and $a_j \in C$ (hence involving no further factor of $p = \partial_1$), so it carries a positive power of some transverse ∂_i , $i > 1$. Such a monomial contributes only to coefficients c_i with $i < N$, never to c_N ; hence c_N receives no contribution besides the coefficient of p^N itself, giving $c_N = 1$ identically, i.e. P is monic of degree N in ξ_1 . Evaluating at $(1, 0, \dots, 0)$ then gives $P(1, 0, \dots, 0) = 1$. The coefficient of $p^N = \partial_1^N$ is 1. Expand each $c_j \in C$ in PBW monomials. A term of $c_j p^j$ contributing order N has transverse derivative order $N - j$ and zero x -degree, because $\deg_B c_j \leq N - j$. For $j < N$ it therefore contains some transverse ∂_i , $i > 1$, and cannot contribute to ξ_1^N . Thus its coefficient in P is 0, so P is monic in ξ_1 and $P(1, 0, \dots, 0) = 1$. $\square$

5. x_1 AS A SURJECTIVE MAP

We define $E = x_1 p$ and

$$(5.1) \quad I = \tilde{d} A_n + x_1^N \tilde{d} A_n, \quad Q = A_n / I.$$

The goal of this subsection will be to show that $Q = Q x_1$ (Proposition 5.3).

Let $A_{\geq 0}$ denote the subring generated by the transverse Weyl algebra, x_1 , and $E = x_1 p$. The argument below uses this concrete subring and PBW monomials; no Euler grading is needed. By the commutation relations of the Weyl algebra, it holds that

$$(5.2) \quad \begin{aligned} E x_1 &= x_1 (E + 1), & x_1 E &= (E - 1) x_1, \\ E p &= p (E - 1), & p E &= (E + 1) p \end{aligned}$$

Additionally, x_1 commutes with the transverse algebra, so, together with $x_1 E = (E - 1) x_1$, x_1 is normal in $A^{\geq 0}$.

Formal comparison: Lean uses the subring generated by the transverse algebra, x_1 and x_1p , and one-sided normality $x_1r = r'x_1$, which is all that is used

AI comment ALG-02. Proposed change — author acceptance pending.
 The displayed products below have reversed shifts. For $E = x_1p$ and $[p, x_1] = 1$, the correct identities are $x_1^m p^m = \prod_{i=0}^{m-1} (E - i)$ and $p^m x_1^m = \prod_{i=1}^m (E + i)$. Already $m = 1$ gives $x_1p = E$, $px_1 = E + 1$. Adjust the induction accordingly; the next residue proof already uses these correct products.

Lemma 5.1. *For every $m \geq 0$,*

$$(5.3) \quad x_1^m p^m = \prod_{i=1}^m (E - i) \prod_{i=0}^{m-1} (E - i), \quad p^m x_1^m = \prod_{i=0}^{m-1} (E + i) \prod_{i=1}^m (E + i),$$

the factors in each product commuting with one another.

Proof. Since both products in (5.3) are polynomials in the single element E , their factors commute. We induct on m ; both identities are trivial for $m = 0$. Assuming the first identity for m , $x_1^{m+1} p^{m+1} = x_1 (\prod_{i=1}^m (E - i)) p = (\prod_{i=1}^m (E - 1 - i)) x_1 p = (\prod_{i=2}^{m+1} (E - i)) (E - 1) = \prod_{i=1}^{m+1} (E - i)$,

$$x_1^{m+1} p^{m+1} = x_1 \left(\prod_{i=0}^{m-1} (E - i) \right) p = \left(\prod_{i=0}^{m-1} (E - 1 - i) \right) x_1 p = \left(\prod_{i=1}^m (E - i) \right) E = \prod_{i=0}^m (E - i),$$

using $x_1 E = (E - 1)x_1$ repeatedly and the definition of E . The second identity follows identically from $Ep = p(E - 1)$:

$$p^{m+1} x_1^{m+1} = p \left(\prod_{i=1}^m (E + i) \right) x_1 = \left(\prod_{i=1}^m (E + 1 + i) \right) p x_1 = \left(\prod_{i=2}^{m+1} (E + i) \right) (E + 1).$$

□

Lemma 5.2. *There exists an element $U \in A_{\geq 0}$ such that*

$$(5.4) \quad 1 + Ux_1 \in I.$$

AI addition: Suggested proof matching the checked route.

Proof. Write $T' = A_{n-1}(k)$ for the transverse algebra and use the subring $R \subseteq A_n$ generated by T' , x_1 and $E = x_1p$. The PBW normal form puts the coefficients on the left: every element is uniquely $\sum_j c_j p^j$ with $c_j \in C = T'[x_1]$, that is, a sum of words $b x_1^a p^j$ with $b \in T'$. The proof has five steps.

(a) R contains $\mathbb{Q}[E]$. It is *one-sidedly normal*: for $r \in R$ there is $r' \in R$ with $x_1 r = r' x_1$. The set of such r is a subring, and it contains the generators, because $x_1 b = b x_1$, $x_1 x_1 = x_1 x_1$ and $x_1 E = (E - 1)x_1$.

(b) The polynomials $\prod_{i=0}^{N-1} (X - i)$ and $\prod_{i=1}^N (X + i)$ have the disjoint root sets $\{0, \dots, N - 1\}$ and $\{-1, \dots, -N\}$. They are therefore coprime in $\mathbb{Q}[X]$. With Lemma 5.1 (corrected form) this gives $u, v \in \mathbb{Q}[X]$ with $p^N x_1^N u(E) + x_1^N p^N v(E) = 1$.

(c) The normal form of Proposition 4.2 gives $\tilde{d} = p^N + \sigma$, where σ is a sum of words $b x_1^a p^j$ with $j < N$. Then

$$\tilde{d} (x_1^N u(E)) + (x_1^N \tilde{d}) v(E) = 1 + \sigma x_1^N u(E) + x_1^N \sigma v(E),$$

and the left side lies in I .

(d) Each word $w = b x_1^a p^j$ with $j < N$ has a right factor x_1 on both sides, with cofactors in R . By Lemma 5.1 and (a),

$$\begin{aligned} w x_1^N &= b x_1^a (p^j x_1^j) x_1^{N-j} = \left[b x_1^a \prod_{i=1}^j (E + i) x_1^{N-j-1} \right] x_1, \\ x_1^N w &= b x_1^{N+a-j-1} x_1 (x_1^j p^j) = \left[b x_1^{N+a-j-1} \prod_{i=0}^{j-1} (E - 1 - i) \right] x_1. \end{aligned}$$

By additivity, $\sigma x_1^N = U' x_1$ and $x_1^N \sigma = V' x_1$ with $U', V' \in R$.

(e) Since $x_1 u(E) = u(E-1)x_1$, the residue equals $U x_1$ with $U = U' u(E-1) + V' v(E-1) \in R$. Hence $1 + U x_1 \in I$. $\square$

Formal comparison: the ring identity is Lean's; the factorization uses left coefficients $c x^a p^j$

AI comment COMP-04. Proposed change — review the red proof. The concrete transverse–Euler subring and the two explicit monomial factorizations replace the grading detour. The displayed calculation follows the checked remainder argument and uses only one-sided normality. [Readable correspondence note.](#)

Proposition 5.3. *Right multiplication by x_1 on Q is surjective.*

AI addition: Suggested proof matching the checked route.

Proof. Let R, U be as in Lemma 5.2, and write $a \equiv b$ for $a - b \in I$. Put

$$\mathcal{P} := \{a \in A_n : \text{for every } r \in R \text{ there is } r' \in R \text{ with } ra \equiv r'\}.$$

$\mathcal{P}$ is a subring containing R . It is closed under products: if $ra \equiv s$ and $sb \equiv t$, then $rab \equiv sb \equiv t$, because I is a right ideal. It also contains p . Indeed, for $r \in R$ pick $r' \in R$ with $x_1 r = r' x_1$. From $(1 + U x_1) r p \in I$ we get $r p \equiv -U x_1 r p = -U r' x_1 p \in R$.

The Ore normal form shows that T', x_1 and p generate A_n , and $T', x_1 \in R$. Hence $\mathcal{P} = A_n$. Given $a \in A_n$, take $r = 1$: then $a \equiv r_a$ for some $r_a \in R$. Choosing r' with $x_1 r_a = r' x_1$ gives $a \equiv r_a \equiv -U x_1 r_a = (-U r') x_1$. So the class of a is the class of $-U r'$ multiplied on the right by x_1 . $\square$

AI comment COMP-05. Proposed change — review the red proof. The direct element argument is shorter and follows the checked subring-closure proof of surjectivity. [Readable correspondence note.](#)

6. VANISHING COKERNEL

In this section and the next, C denotes the commutative symbol ring $k_0[x, \xi] = T[x_1, \xi_1]$ (not the Ore coefficient algebra of Convention 3.1), and Corollary 7.1 of Section 7, whose proof uses only Lemma 4.5, is used freely.

AI comment NOTATION-01. Proposed change — author acceptance pending. C below denotes the commutative symbol ring $T[x_1, \xi_1]$, whereas Convention 3.1 used C for the noncommutative coefficient algebra $A_{n-1}(k)[x_1]$. Please rename one or explicitly announce the change of notation.

Put $M = \text{gr } Q$ and

$$T = k_0[x_2, \dots, x_n, \xi_2, \dots, \xi_n], \quad C = T[x_1, \xi_1],$$

$$V = \ker(R_{x_1} : M \rightarrow M), \quad U = \text{coker}(R_{x_1} : M \rightarrow M) = M/x_1M.$$

With R_{x_1} being the right multiplication ~~with~~^{by} x_1 . If $n = 1$, we define $T = k_0$. We want to prove that $U = 0$ by comparing two lengths over a localization of T .

Definition 6.1 (~~[13, 12]~~[37, Tags 02LY, 00J3, 00IV]). *The length ℓ_T of a ~~finite T -module~~ T -module of finite length M is defined as the number of steps in a composition series of M such that*

$$0 = M_0 \subsetneq M_1 \subsetneq \dots \subsetneq M_{\ell_T} = M$$

where each successive quotient M_i/M_{i-1} is simple (i.e. has no proper nontrivial submodules). The length is well-defined, as it does not depend on the choice of the composition series. Lengths are additive for short exact sequences.

Lemma 6.2. *The modules U and V are finite over T , and $\text{Supp}_T V \subseteq \text{Supp}_T U$. If $\mathfrak{q}$ is a minimal prime in $\text{Supp}_T U$, then $U_{\mathfrak{q}}$ and $V_{\mathfrak{q}}$ have finite length over $T_{\mathfrak{q}}$.*

Proof. The good quotient filtration makes $M = \text{gr } Q$ finitely generated over $C = \text{gr } A_n = k_0[x, \xi] = T[x_1, \xi_1]$. Since P annihilates M (Corollary 7.1), this action factors through $C/(P)$, which, because P is monic of degree N in ξ_1 , is generated over $T[x_1]$ by $1, \xi_1, \dots, \xi_1^{N-1}$. Thus M is finite over $T[x_1]$. Because M is finite, and $T[x_1]$ is a Noetherian ring, the kernel V and cokernel U of multiplication in M by x_1 are also finite over $T[x_1]$. Since they are annihilated by x_1 , they are finite over T .

Suppose $\mathfrak{p}$ is a prime of T where $U_{\mathfrak{p}} = (M/x_1M)_{\mathfrak{p}} = (M_{\mathfrak{p}}/x_1M_{\mathfrak{p}}) = 0$. Consequently, $M_{\mathfrak{p}} = x_1M_{\mathfrak{p}}$, i.e. multiplication by x_1 is a surjective endomorphism of the finite module $M_{\mathfrak{p}}$ over the corresponding localization of C . By ~~Vasconcelos's theorem [21]~~[Theorem 2.4] ~~Vasconcelos' theorem [21, Theorem 2.4]~~, this endomorphism is also injective, which means $V_{\mathfrak{p}} = \ker(x_1 : M_{\mathfrak{p}} \rightarrow M_{\mathfrak{p}}) = 0$, i.e. $\mathfrak{p} \notin \text{Supp}_T(V)$, and therefore $\text{Supp}_T V \subseteq \text{Supp}_T U$.

After localization at a minimal support prime $\mathfrak{q}$ of U , both finite T -modules have support contained in the closed point. They therefore have finite length. $\square$

AI comment ALG-04. Information — earlier request resolved in the proposed text. The finite- T argument is present. The page proof below now constructs its T -action and localizes only the pages.

Lemma 6.3. *For every minimal prime $\mathfrak{q}$ in $\text{Supp}_T U$,*

$$\ell_{T_{\mathfrak{q}}}(U_{\mathfrak{q}}) \leq \ell_{T_{\mathfrak{q}}}(V_{\mathfrak{q}}).$$

AI addition: Suggested proof matching the checked route.

Proof. Only the pages carry a T -module structure; we localize those pages. Put $G^p := F_{-p}Q$ for $p \leq 0$ and $G^p := 0$ for $p > 0$. This is a decreasing, exhaustive filtration with $G^1 = 0$. The complex is the k -linear map $f = R_{x_1}: Q \rightarrow Q$, with $f(G^p) \subseteq G^p$. For $r \geq 0$ define

$$A_r^p = \frac{G^p \cap f^{-1}(G^{p+r})}{G^{p+1} \cap f^{-1}(G^{p+r})}, \quad B_r^p = \frac{G^p}{(G^p \cap f(G^{p-r+1})) + G^{p+1}},$$

and $A_r = \bigoplus_{p \in \mathbb{Z}} A_r^p$, $B_r = \bigoplus_{p \in \mathbb{Z}} B_r^p$. Applying f induces $d_r: A_r^p \rightarrow B_r^{p+r}$: changing a representative by the source denominator changes its image by the target denominator. Thus $A_0 = B_0 = M$ and d_0 is multiplication by x_1 . The proof has four steps.

Step 1: T acts on the pages. Let a be a transverse generator x_j or ∂_j ($j \geq 2$), of order $e_a \in \{0, 1\}$. Then R_a is k -linear, $R_a(G^p) \subseteq G^{p-e_a}$, and R_a commutes with f because $ax_1 = x_1a$. It therefore induces maps $A_r^p \rightarrow A_r^{p-e_a}$ and $B_r^p \rightarrow B_r^{p-e_a}$ that commute with d_r . For two generators, $[R_a, R_b] = \pm R_{[a,b]}$ with $[a, b] \in F_{e_a+e_b-1}$, so it maps G^p into $G^{p-e_a-e_b+1}$. An f -commuting operator with this extra drop induces 0 on every page, because its image lies in the denominator. The induced operators therefore commute. Letting $x_j, \xi_j \in T$ act by R_{x_j} and R_{∂_j} makes every A_r, B_r a T -module and every d_r T -linear. For $r = 0$ this is the T -structure of $M = \text{gr } Q$, so $A_1 \cong V$ and $B_1 \cong U$ as T -modules. In addition, $A_{r+1} \cong \ker d_r$ and $B_{r+1} \cong \text{coker } d_r$ as T -modules. Indeed, if $d_r([z]) = 0$, write $f(z) = f(w) + v$ with $w \in G^{p+1}$ and $v \in G^{p+r+1}$. Then $z - w$ represents the same source class and lies in $G^p \cap f^{-1}G^{p+r+1}$; its kernel of passage back is exactly the denominator of A_{r+1}^p . For the target, quotienting B_r^{p+r} by $d_r(A_r^p)$ enlarges the boundary denominator from $G^{p+r} \cap f(G^{p+1})$ to $G^{p+r} \cap f(G^p)$, giving B_{r+1}^{p+r} . Consequently $0 \rightarrow A_{r+1} \rightarrow A_r \xrightarrow{d_r} B_r \rightarrow B_{r+1} \rightarrow 0$ is exact.

Step 2: exhaustion, before localization. Let $y \in G^p$ represent a class of B_1^p . By Proposition 5.3, $y = zx_1$ for one $z \in Q$, and by exhaustiveness $z \in G^s$ for some s . For $r \geq \max(1, p - s + 1)$ we have $z \in G^{p-r+1}$. Hence $y \in G^p \cap f(G^{p-r+1})$, and the class of y vanishes in B_r^p . In terms of order: if $y \in F_m Q$ and the chosen preimage lies in $F_{m'} Q$, the class dies on page $r = \max(1, m' - m + 1)$. The maps $\beta_r: B_1 \rightarrow B_r$, the identity on representatives, are surjective and T -linear, and their kernels increase with r . An element of B_1 has finitely many homogeneous components, so taking the largest r shows that every $u \in B_1$ has some r with $\beta_r(u) = 0$.

Step 3: localization. Let $S = T \setminus \mathfrak{q}$. By Lemma 6.2, $S^{-1}B_1 \cong U_{\mathfrak{q}}$ has finite length, so it is Noetherian over $T_{\mathfrak{q}}$. The maps $S^{-1}\beta_r$ are surjective with increasing kernels. Each u/s is killed by $S^{-1}\beta_r$ for the r belonging to u . The kernels therefore stabilize at all of $S^{-1}B_1$, and $S^{-1}B_R = 0$ for some R .

Step 4: telescoping. Localizing the exact sequences $0 \rightarrow A_{r+1} \rightarrow A_r \xrightarrow{d_r} B_r \rightarrow B_{r+1} \rightarrow 0$, all modules have finite length because $A_1 \cong V$ and $B_1 \cong U$ do, and subsequent pages are submodules and quotients. Additivity gives $\ell(S^{-1}B_r) + \ell(S^{-1}A_{r+1}) = \ell(S^{-1}A_r) + \ell(S^{-1}B_{r+1})$. Summing for $r = 1, \dots, R-1$ cancels every intermediate term and gives $\ell(U_{\mathfrak{q}}) + \ell(S^{-1}A_R) = \ell(V_{\mathfrak{q}}) + \ell(S^{-1}B_R) = \ell(V_{\mathfrak{q}})$. $\square$

AI comment ALG-09. Information — exhaustion repaired in the proposed proof. A single preimage is chosen before localization; exhaustiveness makes its class die on a late page. No iterated-preimage assertion remains in the selected proof.

AI comment COMP-06. Proposed change — review the red page proof. It supplies the T -action, the single-preimage exhaustion argument, finite-length stabilization after localization, and the telescoping inequality. This is the checked page route. [Readable correspondence note.](#)

On the other hand, we can also establish a different inequality.

Lemma 6.4. *For every minimal prime $\mathfrak{q}$ in $\text{Supp}_T U$,*

$$\ell_{T_{\mathfrak{q}}}(U_{\mathfrak{q}}) > \ell_{T_{\mathfrak{q}}}(V_{\mathfrak{q}}).$$

Proof. By Gabber involutivity (Theorem 3.8) and Lemma 3.9, every minimal prime $\mathfrak{p}$ of $\text{Ann}_C M$ is involutive. It follows, that no $\mathfrak{p}$ contains x_1 : Assume $x_1 \in \mathfrak{p}$. Since $P \in \mathfrak{p}$, also the Poisson brackets

$$\{x_1, P\}, \quad \{x_1, \{x_1, P\}\}, \dots$$

lie in $\mathfrak{p}$. $\{x_1, -\}$ differentiates once with respect to ξ_1 , and because P is homogeneous of degree N and monic in ξ_1 , its N th derivative is the nonzero constant $N!$ - therefore the N th bracket would put $N!(-1)^N N!$ in $\mathfrak{p}$. Since $\mathfrak{p}$ is a proper prime ideal, no nonzero constant can be in $\mathfrak{p}$, so we must conclude $x_1 \notin \mathfrak{p}$. This result holds under localization at $T \setminus \mathfrak{q}$, writing $M' = M_{\mathfrak{q}}$ for the finite module over $C' = (T \setminus \mathfrak{q})^{-1}C$.

~~Consider the increasing chain of kernels $\ker(x_1 : M' \rightarrow M') \subseteq \ker(x_1^2 : M' \rightarrow M') \subseteq \dots$. Since M' is Noetherian, this chain stabilizes at some point m where $K := \ker(x_1^m) = \ker(x_1^{m+1}) = \dots$.~~

AI comment EDIT-02. Proposed change — author acceptance pending. The undefined command `\cots` is aliased to `\cdots` in the annotation preamble so this draft compiles. The kernel-chain argument is repeated immediately below; keep the fuller second version when resolving this comment.

Write $M' = M_{\mathfrak{q}}$, a finite module over $C' = (T \setminus \mathfrak{q})^{-1}C$; this is the same localization used throughout. As shown above, no minimal prime of $\text{Ann}_C(M)$ contains x_1 , and this persists after localizing at $T \setminus \mathfrak{q}$: no minimal prime of

$\text{Ann}_{C'}(M')$ contains x_1 either, **since the annihilator of the finite module M commutes with localization.**

The chain $\ker(x_1) \subseteq \ker(x_1^2) \subseteq \cdots$ of submodules of M' stabilizes, since M' is Noetherian over C' ; let $K := \ker(x_1^m : M' \rightarrow M')$ at a stage where it stabilizes, so $x_1 K \subseteq K$ and $\ker(x_1^{m+1}) = \ker(x_1^m) = K$. Each quotient $\ker(x_1^{i+1})/\ker(x_1^i)$ maps to $V_q = \ker(x_1 : M' \rightarrow M')$ injectively via $v \mapsto x_1^i v$, so, as V_q has finite length, so does each quotient and hence K itself.

Two elementary facts about K :

- $x_1^{-1}K = K$: if $v \in K$ then $x_1 v \in K$ (as $x_1 K \subseteq K$); if $x_1 v \in K = \ker(x_1^m)$ then $x_1^{m+1}v = 0$, so $v \in \ker(x_1^{m+1}) = K$.
- $K \cap x_1 M' = x_1 K$: " $\supseteq$ " is trivial. For " $\subseteq$ ", if $w = x_1 v \in K$ for some $v \in M'$, then $v \in x_1^{-1}K = K$, so $w \in x_1 K$.

For any endomorphism T of a finite-length module N , additivity along $0 \rightarrow \ker T \rightarrow N \rightarrow TN \rightarrow 0$ and $0 \rightarrow TN \rightarrow N \rightarrow N/TN \rightarrow 0$ gives $\ell(\ker T) = \ell(N/TN)$. Applying this to $T = x_1|_K$: since $\ker(x_1) \subseteq K$, we have $\ker(x_1|_K) = \ker(x_1) = V_q$, so

$$\ell(K/x_1 K) = \ell(V_q).$$

The natural surjection $M'/x_1 M' \rightarrow (M'/K)/x_1(M'/K)$ has kernel $(K + x_1 M')/x_1 M' \cong K/(K \cap x_1 M') = K/x_1 K$, giving the exact sequence

$$0 \longrightarrow K/x_1 K \longrightarrow U_q \longrightarrow (M'/K)/x_1(M'/K) \longrightarrow 0,$$

so by additivity and using $\ell(K/x_1 K) = \ell(V_q)$,

$$\ell(U_q) = \ell(V_q) + \ell((M'/K)/x_1(M'/K)).$$

It remains to show the last term is positive.

Since $\mathfrak{q} \in \text{Supp}_T U$, $U_q = M'/x_1 M' \neq 0$; choose $\mathfrak{P} \in \text{Supp}_{C'}(M'/x_1 M')$ and a minimal prime $\mathfrak{p} \subseteq \mathfrak{P}$ of M' . We have shown above that $x_1 \notin \mathfrak{p}$, so x_1 is a unit in the localization at $\mathfrak{p}$; as K is annihilated by a power of x_1 , $K_{\mathfrak{p}} = 0$. From $0 \rightarrow K_{\mathfrak{p}} \rightarrow M'_{\mathfrak{p}} \rightarrow (M'/K)_{\mathfrak{p}} \rightarrow 0$ we get $(M'/K)_{\mathfrak{p}} \cong M'_{\mathfrak{p}} \neq 0$, so $\mathfrak{p} \in \text{Supp}(M'/K)$; since support is closed under passing to larger primes and $\mathfrak{p} \subseteq \mathfrak{P}$, also $\mathfrak{P} \in \text{Supp}(M'/K)$, i.e. $(M'/K)_{\mathfrak{P}} \neq 0$.

Since $\mathfrak{P} \in \text{Supp}_{C'}(M'/x_1 M')$ and $M'/x_1 M'$ is annihilated by x_1 , we have $x_1 \in \mathfrak{P}$. If $x_1(M'/K)_{\mathfrak{P}} = (M'/K)_{\mathfrak{P}}$, then as x_1 lies in the maximal ideal of the local ring $C'_{\mathfrak{P}}$, Nakayama's lemma [22](see [37, Tag 00DV]) would force $(M'/K)_{\mathfrak{P}} = 0$, a contradiction. Hence $((M'/K)/x_1(M'/K))_{\mathfrak{P}} \neq 0$, so $(M'/K)/x_1(M'/K) \neq 0$.

Therefore $\ell(U_q) = \ell(V_q) + \ell((M'/K)/x_1(M'/K)) > \ell(V_q)$. $\square$

AI comment COMP-07. Information — proof comparison; no edit requested. The formal proof uses Gabber directly for minimal primes. If such a prime contained x_1 , the N -fold bracket of the monic symbol P with x_1 would put the unit $\pm N!$ in that prime. The subsequent stable-kernel length computation and Nakayama argument agree with the printed proof. [Mathematical proof \(COMP-07, p. 4\).](#)

Lemma 6.5. *The cokernel $U = M/x_1M$ of right multiplication by x_1 on $M = \text{gr } Q$ vanishes, i.e. $U = 0$.*

Proof. If $U \neq 0$, it has a minimal support prime $\mathfrak{q}$. Lemma 6.2 makes the relevant lengths finite, and Lemmas 6.3 and 6.4 contradict one another. Thus $U = M/x_1M = 0$. $\square$

7. CHARACTERISTICS OF THE CHARACTERISTIC VARIETY

Corollary 7.1. *Equip Q with the order filtration $Q_m := qF_m = \text{im}(F_m \rightarrow Q)$ induced by the cyclic generator q denoting the class of 1 in Q . Then*

$$(7.1) \quad \text{Char}(Q) \subseteq V(P) \subseteq T^*\mathbb{A}^n.$$

Proof. $Q_\bullet$ is a good filtration because Q is cyclic. Since $q\tilde{d} = [\tilde{d}] = 0$ in Q , the symbol $P = \sigma_N(\tilde{d})$ annihilates the class $\bar{q} \in Q_0/Q_{-1}$ of q in $\text{gr } Q$: the element $q\tilde{d} \in Q_N$ is zero, and its image in Q_N/Q_{N-1} is $\bar{q} \cdot P$. Since $\text{gr } Q$ is generated over $k_0[x, \xi]$ by $\bar{q}$, we get $P \in \text{Ann}_{k[x, \xi]} \text{gr } Q$, and $\text{Char}(Q) = V(\text{Ann } \text{gr } Q) \subseteq V(P)$ because V reverses inclusions. $\square$

Proposition 7.2. *The canonical quotient satisfies*

$$(7.2) \quad \text{Char}(Q) \cap \pi^{-1}(H) = \emptyset,$$

where $\pi : T^*\mathbb{A}^n \rightarrow \mathbb{A}^n$ is the cotangent projection; equivalently, $\text{Ann}_C \text{gr } Q + (x_1) = C$, i.e. no prime of $\text{Supp}_C \text{gr } Q$ contains x_1 .

Proof. By Lemma 6.5 $U = M/x_1M = 0$. Since M is finite over C , its quotient by x_1 has support $\text{Supp}_C(M) \cap V(x_1)$. Lastly, $\text{Supp}_C(M) = \text{Char}(Q)$, and $V(x_1)$ is the set of all points $\in k^{2n}$ where $x_1 = 0$, which is nothing else than the pre-image of $H = \{x_1 = 0\}$ under the projection, i.e. $V(x_1) = \pi^{-1}(H)$. Since $M/x_1M = 0$, it has no support, so $\text{Supp}(M/x_1M) = \text{Supp}_C(M) \cap V(x_1) = \text{Char}(Q) \cap \pi^{-1}(H) = \emptyset$.

Here points are prime ideals of C : the argument is $\text{Supp}_C(M/x_1M) = \text{Supp}_C M \cap V(x_1)$ for the finite C -module M . $\square$

Formal comparison: stated for the support transported along $\xi \mapsto -\xi$, which fixes x_1 ; transported back verbatim

AI comment COMP-08. Information — proof comparison; no edit requested. The formal conclusion is scheme-theoretic: $\text{Supp}(\text{gr } Q) \cap V(x_1) = \emptyset$, equivalently $\text{Ann}(\text{gr } Q) + (x_1) = (1)$. It therefore persists under field extension. The formal statement uses the support transported by $\xi \mapsto -\xi$; this fixes x_1 , so the avoidance statements are equivalent. [Mathematical proof \(COMP-08, p. 5\)](#).

8. A CONORMAL DIRECTION AT INFINITY

Throughout this section and the next, $\mathbb{C}$ may be replaced by any algebraically closed field K of characteristic zero; the formal statements are proved in that generality, and Section 10 uses $K = \bar{k}$. Let K be an algebraically closed field of characteristic zero and $X = \mathbb{A}_K^n$. Because $\mathbb{A}_K^n$ has a global coordinate frame, $T^*X = X \times (K^n)^*$ holds and we can therefore identify all cotangent fibres with $(K^n)^*$. We write $[Z_0 : \cdots : Z_n]$ for homogeneous coordinates on the projective space $\mathbb{P}^n$, and embed $X \hookrightarrow \mathbb{P}^n$ as the affine chart $\{Z_0 \neq 0\}$ via $[1 : x_1 : \cdots : x_n]$, so that $x_i = Z_i/Z_0$ and the hyperplane at infinity is $\{Z_0 = 0\}$. Let $Y \subset X$ be an irreducible closed subvariety of X . We define $\text{Dir}(Y)$ to be the Zariski closure in $\mathbb{P}((K^n)^*)$ of the nonzero conormal vectors at all smooth affine points of Y projected onto $\mathbb{P}((K^n)^*)$.

Formal comparison: Y given by a prime ideal; conormal space = annihilator of the Zariski tangent space; closure by homogeneous polynomials

Lemma 8.1. *Suppose a formal arc in the projective closure of Y has generic point in the smooth affine locus, and put $m = \dim Y$. Let $L \subset \mathbb{C}K[[t]]^{n+1}$ be a direct summand of rank $m + 1$ whose generic fibre is the affine cone over the embedded projective tangent m -plane. If the reduction L_0 satisfies $L_0 \subset \{Z_1 = 0\}$, then $[dx_1] \in \text{Dir}(Y)$.*

Proof. To prove this lemma, we will construct an arc such that its generic fibre is a conormal direction at a smooth point of Y (hence lies in $\text{Dir}(Y)$), identify its special fibre as $[dx_1]$ and conclude from the closedness of $\text{Dir}(Y)$ that it must also lie in $\text{Dir}(Y)$. The annihilator $L^\perp$ of L is also a direct summand, and therefore reduces onto $L_0^\perp$. The dual basis vector $Z_1^\vee$ annihilates L_0 since $L_0 \subset \{Z_1 = 0\} = \ker(Z_1^\vee)$, and hence $Z_1^\vee \in L_0^\perp$. Because the reduction is a surjective map, we can lift $Z_1^\vee \in L_0^\perp$ to $\ell(t) \in L^\perp$, satisfying $\ell(0) = Z_1^\vee$. At the generic point, $\ell(t)$ annihilates the generic fibre of L and hence by hypothesis the cone over the embedded projective tangent space. The fibre coordinates $(\ell_1(t), \dots, \ell_n(t))$ of $\ell(t)$ in the affine chart $Z_0 \neq 0$ therefore form a nonzero affine conormal covector. Because $\ell(0) = Z_1^\vee \neq 0$, the reduction of the fibre coordinates at $t = 0$ is nonzero, so their projective direction extends over $\mathbb{C}K[[t]]$. Specifically, this reduction is $(1, 0, \dots, 0)$, giving the special fibre $[dx_1]$. Since $\text{Dir}(Y) \subset \mathbb{P}((\mathbb{C}^n)^*)$ is closed, hence proper over $\mathbb{C}$, $[dx_1]$ must lie in $\text{Dir}(Y)$ by the valuative criterion [16, Theorem II.4.7]. The generic direction lies in $\text{Dir}(Y)$: every homogeneous polynomial vanishing on $\text{Dir}(Y)$ vanishes on the conormal directions of the $\mathbb{C}K((t))$ -point $\gamma(\eta)$ as well (the conormal variety over Y_{sm} is defined over $\mathbb{C}K$, and its $\mathbb{C}K$ -points are dense in it). Hence the defining equations of the closed set $\text{Dir}(Y)$ vanish on the direction of $\ell(t)$ in $\mathbb{C}K((t))$, hence on its value at $t = 0$, which is $[dx_1]$. No valuative criterion is needed. $\square$

AI comment GEO-04. Proposed change — author acceptance pending. The printed proof did not justify that a conormal direction at the $\underline{\mathbb{C}}K((t))$ -point $\gamma(\eta)$ belongs to $\text{Dir}(Y)$, which is defined from $\underline{\mathbb{C}}K$ -points; the red text supplies the scalar-extension step, which the formalization proves as `scalarExtension_vanishing`.

Formal comparison: any algebraically closed field and any axis; auxiliary, not used by the formal proof of Theorem 8.2

Formal comparison: generic point only Zariski-tangent; no smoothness needed for this step

AI comment COMP-10. Information — separately checked auxiliary lemma. Its stated tangent-lattice assumptions match the formal criterion. The selected proof of the following theorem uses the Laurent route and does not require constructing such a lattice. [Readable correspondence note](#).

Theorem 8.2. *If Y is nonempty and $Y \cap H = \emptyset$, then*

$$[dx_1] \in \text{Dir}(Y).$$

Formal comparison: over every algebraically closed field of characteristic zero; the selected proof below follows the checked Laurent route

AI comment ROUTE-01. Proposed change — selected geometric proof for review. The red proof below is the already checked divisorial/Laurent argument. It avoids the unfinished bridge from a normalized projective chart to a tangent lattice. The prior argument is retained verbatim in marked source-history comments; Max reviews this selected proof, not that historical route.

AI addition: Suggested proof matching the checked route.

Proof. Let K be algebraically closed of characteristic zero, let $\mathfrak{p} \subset K[x_1, \dots, x_n]$ with $n \geq 1$ be a prime ideal, and let $Y = V(\mathfrak{p}) \subset K^n$ satisfy $Y \cap \{x_1 = 0\} = \emptyset$. For a field $L \supseteq K$ and $y \in L^n$ write $\nabla f(y) = (\partial_1 f(y), \dots, \partial_n f(y))$, and let

$$\mathcal{C}_L = \{(y, \xi) \in L^n \times L^n : f(y) = 0 \ \forall f \in \mathfrak{p}, \ \xi \in \text{span}_L\{\nabla f(y) : f \in \mathfrak{p}\}\}$$

be the *equation-conormal locus* over L (singular y allowed). Since $\mathfrak{p} = I(Y)$, the fibre of $\mathcal{C}_K$ over $y \in Y_{\text{sm}}$ is $N_y^* Y$, with covectors written in the basis dx_i . Put $N(Y) = \bigcup_{y \in Y_{\text{sm}}} N_y^* Y \subset K^n$. We prove: (i) every $P \in K[\xi_1, \dots, \xi_n]$ vanishing on $N(Y)$ satisfies $P(e_1) = 0$; (ii) $[dx_1] \in \text{Dir}(Y)$. For $K = \mathbb{C}$ and Y irreducible, $\mathfrak{p} = I(Y)$ is prime and (ii) is Theorem 8.2. The hypothesis “ Y nonempty” is automatic, since $\mathfrak{p} \neq (1)$. Let $R = K[x]/\mathfrak{p}$, $F = \text{Frac } R$, and let $\bar{x}_i \in R$ be the coordinate functions.

Step 1. (*Polynomial inverse.*) $V(\mathfrak{p} + (x_1)) = \emptyset$, so by the Nullstellensatz there is $g \in K[x]$ with $x_1 g - 1 \in \mathfrak{p}$.

Step 2. (*Constant case.*) If $\bar{x}_1$ is algebraic over K , then $\bar{x}_1 = c \in K$ because K is algebraically closed and R is a domain; so $x_1 - c \in \mathfrak{p}$. For any $y \in Y$ (nonempty by the Nullstellensatz), $(y, e_1) = (y, \nabla(x_1 -$

$c(y)) \in \mathcal{C}_K$. Take $\kappa = K$, constant series $y(t) = y$, $\xi(t) = e_1$, and go to Step 13. This includes $\dim Y = 0$.

- Step 3.** (*Valuation ring and chart.*) Now let $\bar{x}_1$ be transcendental. The ideal $\bar{x}_1 K[\bar{x}_1]$ is proper, so Chevalley's extension theorem gives a valuation ring $W \subset F$ with $K[\bar{x}_1] \subset W$ and $\bar{x}_1 \in \mathfrak{m}_W$. Let $Y_\bullet = (1, \bar{x}_1, \dots, \bar{x}_n)$. Pick j with $Y_a/Y_j \in W$ for all a and put $c_a = Y_a/Y_j$. Let A be the integral closure in F of $C_0 = K[c_0, \dots, c_n, \bar{x}_1]$, whose fraction field is F since $\bar{x}_a = c_a/c_0$. Then A is a finitely generated normal K -domain, finite over C_0 . It lies in W because W is integrally closed, so $\bar{x}_1 \in A$ and $\bar{x}_1^{-1} \notin A$. No projective closure, properness or global normalization is used.
- Step 4.** (*Divisorial DVR.*) Since $\bar{x}_1$ is a nonzero nonunit of A , Krull's principal ideal theorem gives a height-one prime $\mathfrak{q} \ni \bar{x}_1$. As A is Noetherian and normal, $V := A_{\mathfrak{q}}$ is a DVR with $\bar{x}_1 \in \mathfrak{m}_V$. Its residue field $\kappa = \text{Frac}(A/\mathfrak{q})$ is the function field of the prime divisor $D = V(\mathfrak{q})$. Moreover $\Omega_{V/K}$ is a finite V -module. Fix a uniformizer t .
- Step 5.** (*D lies at infinity.*) Divide $Y_\bullet$ by an entry of minimal valuation. This gives $q = (q_0, \dots, q_n) \in V^{n+1}$ with $q_{j_0} = 1$, $\bar{x}_a = q_a/q_0$ and $q_1 = q_0 \bar{x}_1$. If q_0 were a unit, all $\bar{x}_a$ would lie in V . Then $\bar{x}_1^{-1} = g(\bar{x}) \in V$ by Step 1, contradicting $\bar{x}_1 \in \mathfrak{m}_V$. Hence $q_0 \in \mathfrak{m}_V$ (Z_0 vanishes on D) and $j_0 \neq 0$.
- Step 6.** (*Order gap.*) Write $q_0 = t^a u_0$ and $\bar{x}_1 = t^e u$ with $u_0, u \in V^\times$ and $a, e \geq 1$. Then $q_1 = t^{a+e} u_1$ with $u_1 = u_0 u$, which has $b = a + e > a > 0$; explicitly,

$$(8.1) \quad q_0 = t^a u_0, \quad q_1 = t^b u_1, \quad b > a > 0.$$

- Step 7.** (*Kähler visibility.*) $A/\mathfrak{q}$ is integral over the image of $K[c, \bar{x}_1]$, and the residue of $\bar{x}_1$ is 0. So κ is algebraic over $K(\bar{c}_0, \dots, \bar{c}_n)$. The columns q and c are proportional and both have an entry 1, so $q = \mu c$ with $\mu \in V^\times$. Using $\bar{q}_0 = 0$, this gives $K(\bar{c}_0, \dots, \bar{c}_n) = K(\bar{q}_1, \dots, \bar{q}_n)$. The extension is separable because $\text{char } K = 0$, so the $d\bar{q}_j$ span $\Omega_{\kappa/K}$. The exact sequence $\mathfrak{m}_V/\mathfrak{m}_V^2 \rightarrow \kappa \otimes_V \Omega_{V/K} \rightarrow \Omega_{\kappa/K} \rightarrow 0$ with $\mathfrak{m}_V = tV$ then shows that the finite V -module $M := \text{im}(\Omega_{V/K} \rightarrow \Omega_{F/K})$ satisfies

$$M = V dt + \sum_{j=1}^n V dq_j + tM.$$

- Step 8.** (*Nakayama.*) For $P \in V$ put $\sigma(P) = t^{a+1} u_0^2 d(P/q_0) = u_0 t dP - au_0 P dt - tP du_0$, and let $S = \sum_{j=1}^n V \sigma(q_j)$. Since $q_{j_0} = 1$, $\sigma(q_{j_0}) = -au_0 dt - t du_0$. Hence $t dq_j = u_0^{-1}(\sigma(q_j) - q_j \sigma(q_{j_0})) \in S$ and $au_0 t dt = -t\sigma(q_{j_0}) - t^2 du_0 \in S + t^2 M$, where $au_0 \in V^\times$ because $\text{char } K = 0$. By Step 7, $tM \subset S + t(tM)$, and Nakayama gives $tM \subset S$. From $q_1 = t^{a+e} u_1$ we get $\sigma(q_1) = t^{a+e-1} \cdot t\omega_0$ with $\omega_0 = u_0 t du_1 + eu_0 u_1 dt - tu_1 du_0 \in M$. Thus $\sigma(q_1) = \sum_j t^{a+e-1} b_j \sigma(q_j)$

with $b_j \in V$. Dividing by $t^{a+1}u_0^2$ gives

$$d\bar{x}_1 = \sum_{j=1}^n c_j d\bar{x}_j \text{ in } \Omega_{F/K}, \quad c_j = t^{a+e-1}b_j \in \mathfrak{m}_V.$$

- Step 9.** (*Generic conormal covector.*) Let $\xi^F = e_1 - \sum_j c_j e_j \in V^n$, so $\sum_i \xi_i^F d\bar{x}_i = 0$. Localizing the conormal sequence of $K[x] \rightarrow R$ at (0) gives the exact sequence $\mathfrak{p} \otimes F \rightarrow F^n \rightarrow \Omega_{F/K} \rightarrow 0$, where $f \mapsto \nabla f(\bar{x})$. Hence $(\bar{x}, \xi^F) \in \mathcal{C}_F$.
- Step 10.** (*Completion and Laurent point.*) Lift a Noether normalization of $A/\mathfrak{q}$ to $a_1, \dots, a_s \in A$. Every nonzero polynomial in the a_i has nonzero residue, so $E = K(a_1, \dots, a_s) \subset V$, and κ/E is finite and separable. By Hensel's lemma and Cohen's theorem, $\widehat{V} \cong \kappa[[t]]$ as E -algebras. This yields an injective $\varphi : V \rightarrow \kappa[[t]]$ over K with $\varphi(v)(0) = \bar{v}$. It extends to a field embedding $\ell : F \rightarrow \kappa((t))$ over K . Such embeddings preserve $\mathcal{C}$, because $\mathfrak{p} \subset K[x]$. So $y(t) := \ell(\bar{x}) = (\varphi(q_a)/\varphi(q_0))_a$ and $\xi(t) := \varphi(\xi^F) \in \kappa[[t]]^n$ satisfy $(y(t), \xi(t)) \in \mathcal{C}_{\kappa((t))}$ and $\xi(0) = e_1$, since $c_j \in \mathfrak{m}_V$. The base point has a pole, because $\text{ord } \varphi(q_0) = a > 0$. The field $\kappa = \kappa(D)$ has transcendence degree $\dim Y - 1$ over K . No closed point of D is chosen, and the arc's generic point need not be smooth.
- Step 11.** (*Equation- versus smooth-conormal closure.*) Let $P \in K[\xi]$ vanish on $N(Y)$. Since K is perfect, generic smoothness gives $f \notin \mathfrak{p}$ with $Y \cap \{f \neq 0\} \subset Y_{\text{sm}}$. Take $(y, \xi) \in \mathcal{C}_K$ and write $\xi = \sum \lambda_a \nabla h_a(y) = \nabla h(y)$ with $h = \sum \lambda_a h_a \in \mathfrak{p}$. The polynomial $G(x) = P(\nabla h(x))$ vanishes on $Y \cap \{f \neq 0\}$. So $fG \in I(Y) = \mathfrak{p}$, hence $G \in \mathfrak{p}$ and $P(\xi) = G(y) = 0$.
- Step 12.** (*Scalar extension.*) Let $L \supseteq K$ and $(y, \xi) \in \mathcal{C}_L$. Since $d(gh)(y) = g(y)\nabla h(y)$ for $h \in \mathfrak{p}$, we can write $\xi = \sum_{a=1}^r \lambda_a \nabla h_a(y)$ with $h_a \in \mathfrak{p}$ and $\lambda \in L^r$. Put $G(x, \lambda) = P(\sum_a \lambda_a \nabla h_a(x)) \in K[x, \lambda]$. By Step 11, G vanishes at every K -point of $Y \times K^r$. The Nullstellensatz gives $G^N \in \mathfrak{p}K[x, \lambda]$, and evaluating at (y, λ) gives $P(\xi) = 0$.
- Step 13.** (*Evaluation at $t = 0$.*) Apply Step 12 with $L = \kappa((t))$ to the point of Step 2 or Step 10. Then $P(\xi(t)) = 0$. Its constant term is $P(\xi(0)) = P(e_1)$, so $P(e_1) = 0$, which is (i). For (ii), let P be homogeneous and vanish at ξ for every direction $[\xi]$ with $\xi \in N(Y) \setminus 0$. Then $\xi_1 P$ vanishes on $N(Y)$, and (i) gives $P(e_1) = 0$. Therefore $[dx_1] \in \text{Dir}(Y)$.

Characteristic zero is used in Steps 7, 8, 10 and 11. Algebraic closedness is used in Steps 1, 2, 11 and 12. This completes the proof. $\square$

AI comment COMP-11. Information — selected proof correspondence.

The printed red proof now follows the checked Laurent route: a divisorial order gap and a differential relation produce a conormal covector tending to dx_1 , and scalar-extension vanishing proves membership in the ground-field direction closure. [Readable correspondence note.](#)

Formal comparison: any algebraically closed field of characteristic 0; (i) is the affine-cone form used in Section 9

9. THE INVOLUTIVE OBSTRUCTION

Definition 9.1 (Fibre-conical). *A closed subset $W \subseteq T^*X$ is fibre-conical if it is stable under the fibrewise scaling $(y, \xi) \mapsto (y, t\xi)$, $t \in \mathbb{G}_m$.*

Theorem 9.2 (Involutive exclusion). *Let $W \subset T^*X$ be a reduced closed fibre-conical and involutive subset. Let $P \in \mathbb{C}K[\xi_1, \dots, \xi_n]$ be homogeneous and suppose $P(1, 0, \dots, 0) \neq 0$. There is no nonempty W satisfying*

$$W \cap \pi^{-1}(H) = \emptyset, \quad W \subset V(P).$$

AI addition: Formal statement of Theorem 9.2.

Lean proves the following strengthening. Let K be an algebraically closed field of characteristic zero and $n \geq 1$. Let $W \subset K^n \times K^n$ be nonempty and Zariski closed, with $(y, a\xi) \in W$ whenever $(y, \xi) \in W$ and $a \in K^\times$, and with $\{I(W), I(W)\} \subseteq I(W)$. Let $P \in K[\xi_1, \dots, \xi_n]$ be any polynomial, not necessarily homogeneous, with $P(\xi) = 0$ for all $(y, \xi) \in W$ and $P(1, 0, \dots, 0) \neq 0$. Then some $(y, \xi) \in W$ has $y_1 = 0$, i.e. $W \cap \pi^{-1}(H) \neq \emptyset$. The proof uses only the base-relative fragment $\{f \circ \pi, I(W)\} \subseteq I(W)$ for $f \circ \pi \in I(W)$. The Lean bracket is the negative of the one in the paper, which does not affect involutivity. “Involutive” means $\{I(W), I(W)\} \subseteq I(W)$, not that $I(W)$ is a Poisson ideal. For $K = \mathbb{C}$ this contains Theorem 9.2.

Formal comparison: any algebraically closed field of characteristic 0; P need not be homogeneous

AI addition: Suggested proof matching the checked route.

Proof. Work over the algebraically closed field K and suppose $W \cap \pi^{-1}(H) = \emptyset$. Put $J = I(W)$, which is radical, and $B = J \cap K[x]$.

Step 1. (*ξ -homogeneity.*) Write $g \in J$ as $g = \sum_d g_d$ with g_d homogeneous of degree d in ξ . For $(y, \xi) \in W$, the polynomial $a \mapsto \sum_d a^d g_d(y, \xi)$ vanishes on the infinite set $K^\times$. So every $g_d \in J$, i.e. J is ξ -homogeneous.

Step 2. (*Zero section.*) Let $Y = V(B)$. For $y \in Y$ and $g \in J$, $g(y, 0) = g_0(y) = 0$ because $g_0 \in J$ has ξ -degree 0, i.e. $g_0 \in B$. So $Y \times \{0\} \subset V(J) = W$. Hence $\pi(W) = Y$ is closed, and $Y \cap H = \emptyset$.

Step 3. (*Vertical translation.*) Let $f \in B$. Then $f \circ \pi \in J$, so $H_f g := \sum_i \partial_{x_i} f \partial_{\xi_i} g = \pm \{f, g\}$ lies in J for every $g \in J$. This is the only use of involutivity. Fix $y \in Y$ and $g \in J$ and let $\varphi(t) = g(y, t df_y) \in K[t]$. The chain rule gives $\varphi^{(m)}(t) = (H_f^m g)(y, t df_y)$, and $H_f^m g \in J$, so $\varphi^{(m)}(0) = 0$ by Step 2. As $m! \neq 0$ in K , we get $\varphi = 0$, so $(y, t df_y) \in W$ for all t . Since $\sum_a c_a d(f_a)_y = d(\sum_a c_a f_a)_y$ and $\sum_a c_a f_a \in B$, it follows that $\{y\} \times \text{span}\{df_y : f \in B\} \subset W$. No exponential series or composition of flows is needed.

Step 4. (*Component containment.*) Let $\mathfrak{q}$ be a minimal prime of B and $Y_0 = V(\mathfrak{q})$. Taking a product of elements of the other minimal primes that lie outside $\mathfrak{q}$ gives $s \notin \mathfrak{q}$ with $s\mathfrak{q} \subset B$. For $y \in Y_0$ with $s(y) \neq 0$ and $h \in \mathfrak{q}$, we have $d(sh)_y = s(y)dh_y$, so $(y, dh_y) \in W$ by Step 3. For $g \in J$, the polynomial $G(x) = g(x, \nabla h(x))$ therefore vanishes on $Y_0 \cap \{s \neq 0\}$. Hence $sG \in \mathfrak{q}$, so $G \in \mathfrak{q}$, and $(y, \nabla h(y)) \in W$ for every $y \in Y_0$. Absorbing linear combinations into h gives $\{y\} \times \text{span}\{\nabla h(y) : h \in \mathfrak{q}\} \subset W$ for all $y \in Y_0$. In particular $N_y^*Y_0 \subset W_y$ at every smooth point y of Y_0 ,

Step 5. (*Conclusion in the affine cone.*) Y_0 is irreducible, nonempty and disjoint from H . By Step 4 and $W \subset V(P)$, P vanishes on $N(Y_0) = \bigcup_{y \in (Y_0)_{\text{sm}}} N_y^*Y_0 \subset K^n$. By part (i) of Theorem 8.2 in the affine-cone form proved above, $P(1, 0, \dots, 0) = 0$, a contradiction.

Part (i) is a statement about the vanishing ideal of the cone $N(Y_0) \subset K^n$. So $[dx_1]$ never has to be a fibre direction of W over a finite base point, so no direction over a finite base point is asserted. Homogeneity of P is not used, and one component Y_0 suffices. $\square$

AI comment COMP-13. Proposed change — review the red exclusion proof. Finite Taylor identities give conormal containment for a minimal base component. The affine conormal closure then contradicts $P(e_1) \neq 0$. This follows the checked set-level exclusion proof; no formal flow or finite-base-point limit is required. [Readable correspondence note.](#)

AI comment GEO-05. Optional refinement — not required. Optional statement refinements suggested by the formal version, for the authors to decide: in Section 8 say that Y is reduced and irreducible (so $I(Y)$ is prime) and identify $(\mathbb{C}^n)^*$ with $\mathbb{C}^n$ via dx_i ; in Lemma 8.1 the arc is $\mathbb{C}[[t]]$ -valued with affine generic point, and “smooth” can be weakened to “Zariski tangent space”; in Theorem 8.2 “ Y nonempty” is automatic, and the affine-cone form (i) of the red proof is what Theorem 9.2 actually uses; in Theorem 9.2 “reduced” is vacuous for a point set, the last sentence should read “if $W \neq \emptyset$ and $W \subset V(P)$, then $W \cap \pi^{-1}(H) \neq \emptyset$ ”, and homogeneity of P can be dropped.

10. PROOF OF THEOREM 1.1

AI addition: Suggested proof matching the checked route.

Proof. Work over k , pass to $\bar{k}$ for the geometry, and descend the order initial ideal.

Step 1: degenerate cases. For $n = 0$, $A_0(k) = k$ and $d \neq 0$ is a unit, so $R = d^{-1}$ and $F = S = 0$ work. Let $n \geq 1$ and $N = \deg_B d$. If $N = 0$, then $d = c \in k^\times$, and $\ell = x_1$, $F = \ell^0 = 1$, $R = c^{-1}$, $S = 0$ work.

Step 2: normal form. For $N \geq 1$, Proposition 4.2 gives symplectic M, M' with $MM' = M'M = 1$ and $c \in k^\times$ such that $\tilde{d} = c^{-1}\Phi_M(d) \in A_n(k)$ has the normal form of (4.1). Let $I = \tilde{d}A_n + x_1^N \tilde{d}A_n$ and $Q = A_n/I$. It suffices to show $\text{in}(I) = (1)$ (Lemma 10.1).

Step 3: vanishing over $\bar{k}$. The image of $\tilde{d}$ in $A_n(\bar{k})$ is again PBW-monic, so Sections 5–7 apply to $Q_{\bar{k}} = A_n(\bar{k})/I_{\bar{k}}$. Suppose $J := \sqrt{\text{in}(I_{\bar{k}})} \neq (1)$, and let $W = V(J) \subseteq \bar{k}^{2n}$. Then:

- $W \neq \emptyset$ by the Nullstellensatz, and $I(W) = J$.
- W is closed and fibre-conical. The generators $\sigma_L(z)$ of $\text{in}(I_{\bar{k}})$ are homogeneous of degree L in ξ , so $\text{in}(I_{\bar{k}})$ and J are ξ -homogeneous. For $f \in J$ each ξ -homogeneous component f_j lies in J , and $f(y, t\xi) = \sum_j t^j f_j(y, \xi)$.
- W is involutive: $J = \sqrt{\text{Ann gr } Q_{\bar{k}}}$ is involutive by Theorem 3.8 (radical form, a corollary of the minimal-prime form).
- $P = \sigma(\tilde{d}) \in \text{in}(I_{\bar{k}}) \subseteq J$, so $W \subseteq V(P)$, and $P(1, 0, \dots, 0) = 1$ by Lemma 4.5.
- $W \cap \pi^{-1}(H) = \emptyset$ by Proposition 7.2 over $\bar{k}$, i.e. $\text{in}(I_{\bar{k}}) + (x_1) = (1)$.

Theorem 9.2 gives a point of W with $x_1 = 0$, a contradiction. Hence $\text{in}(I_{\bar{k}}) = (1)$.

Step 4: descent. By Lemma 10.1, $\text{in}(I)\bar{k}[x, \xi] = (1)$, hence $\text{in}(I) = (1)$.

Step 5: the certificate over k . By Lemma 10.1, $\text{in}(I) = (1)$ gives $Q = 0$, i.e. $1 \in I$. So $1 = \tilde{d}R + x_1^N \tilde{d}S$ with $R, S \in A_n(k)$.

Step 6: transport back. Since c^{-1} is central, we get

$$1 = \Phi_M(d)(c^{-1}R) + x_1^N \Phi_M(d)(c^{-1}S).$$

Applying $\Phi_M^{-1} = \Phi_{M'}$ gives $1 = dR' + \ell^N dS'$ with $R' = \Phi_{M'}(c^{-1}R)$, $S' = \Phi_{M'}(c^{-1}S)$ and $\ell = \Phi_{M'}(x_1) = \sum_j M'_{1j} z_j$, where row 1 is the row of x_1 and $N = \deg_B d$. $\square$

AI addition: Base change of the order initial ideal.

Lemma 10.1. *Let K/k be a field extension, $I = \tilde{d}A_n(k) + x_1^N \tilde{d}A_n(k)$ and $I_K = \tilde{d}A_n(K) + x_1^N \tilde{d}A_n(K)$. Then $\text{in}(I_K) = \text{in}(I) \cdot K[x, \xi]$. Consequently $\text{Char}(Q_K)$ is the preimage of $\text{Char}(Q)$ under $\text{Spec } K[x, \xi] \rightarrow \text{Spec } k[x, \xi]$. Moreover $Q = 0 \iff \text{in}(I) = (1) \iff \text{in}(I_{\bar{k}}) = (1)$.*

Proof. Identify $A_n(K) = A_n(k) \otimes_k K$ through the PBW basis. Then $F_L(A_n(K)) = F_L \otimes_k K$ and $I_K = I \otimes_k K$ as K -subspaces. Since K is flat over k , $(I \otimes_k K) \cap (F_L \otimes_k K) = (I \cap F_L) \otimes_k K$. Hence every $z \in I_K \cap F_L(A_n(K))$ is a sum $\sum_i c_i w_i$ with $c_i \in K$ and $w_i \in I \cap F_L$. Its order- L symbol is $\sum_i c_i \sigma_L(w_i) \in \text{in}(I)K[x, \xi]$. The reverse inclusion holds because σ_L commutes with the scalar extension.

For the last claim: if $\text{in}(I) = (1)$, then $\text{gr } Q = 0$. Since $F_{-1}Q = 0$ and the filtration is exhaustive, induction on m gives $F_m Q = 0$ for all m , so $Q = 0$. The converse is clear. Finally, a proper ideal of $k[x, \xi]$ has a zero in $\bar{k}^{2n}$ (Nullstellensatz), so it stays proper in $\bar{k}[x, \xi]$. $\square$

Formal comparison: stated for the canonical ideal; the argument uses nothing else

AI comment COMP-15. Proposed change — review the red main proof. Work over k , apply the geometry over $\bar{k}$, descend the canonical order initial ideal, and transport the certificate back. This removes the mixed $k_0/\mathbb{C}$ passage and follows the checked main assembly. [Readable correspondence note.](#)

~~Latly~~**Lastly**, as Stafford already showed, from $1 = dR + FdS$ follows cyclicity of the torsion right module:

Corollary 10.2. *Every finitely generated torsion right $A_n(k)$ -module is cyclic.*

AI comment ALG-06. Proposed change — author acceptance pending. State the consequence for *finitely generated* torsion right modules, as required by Stafford’s two-generator theorem. Write $\phi(r, s) = t_1r + t_2s$. The map ϕ is surjective, not an isomorphism. Correct the final quotient to $T \cong (A_n \oplus A_n)/M$. The displayed right-module generator calculation itself is valid.

AI addition: Suggested proof matching the checked route.

Proof. No two-generator theorem and no Ore condition are needed. Let T be a finitely generated torsion right module over a ring A without zero divisors in which every $d \neq 0$ admits $1 = dR + FdS$; $A = A_n(k)$ qualifies by Theorem 1.1 and because $A_n(k)$ is a domain (the Bernstein symbol of a product is the product of the symbols, in the polynomial ring $k[x, \xi]$).

Two generators. Let $t_1, t_2 \in T$. Choose $d_2 \neq 0$ with $t_2d_2 = 0$ and $e \neq 0$ with $(t_1d_2)e = 0$; then $d := d_2e \neq 0$ annihilates t_1 and t_2 . Write $1 = dR + FdS$ and put $g := t_1 - t_2F$. Then $t_2 = t_2(dR + FdS) = t_2FdS$ and $gdS = t_1dS - t_2FdS = -t_2$, so $t_2 \in gA$ and $t_1 = g + t_2F \in gA$; conversely $g \in t_1A + t_2A$. Hence $t_1A + t_2A = gA$.

Induction. If $t_1, \dots, t_m$ generate T , replace t_{m-1}, t_m by a single generator of $t_{m-1}A + t_mA$ and repeat; after $m - 1$ steps one generator remains. $\square$

AI comment COMP-16. Proposed change — review the red cyclicity proof. The generator $t_1 - t_2F$ is unchanged. Induction on a finite generating set removes the separate two-generator theorem and Ore step. Its certificate calculation also has a standalone checked supplementary statement. Stafford defines stable m -generation on [30, p. 435]; the corollary here deliberately asserts ordinary cyclicity. [Readable correspondence note.](#)

11. OUTLOOK

~~We have proved~~ The cofactors R, S are obtained non-constructively: the argument proves $Q = 0$ without producing a certificate. Effective bounds for $\deg_B R$ and $\deg_B S$ in terms of $\deg_B d$, and an algorithm in the spirit of the constructive two-generator methods, remain open. The Global Stafford extension raises the same questions for rings of differential operators on smooth affine varieties.

AI comment EDIT-01. Proposed change — review the red outlook. The unfinished fragment has been replaced by the open questions on effective certificates and the global extension.

REFERENCES

1. Christopher Albert, *AlgebraicAnalysis v0.3.0*, Zenodo software release v0.3.0, 2026, DOI: <https://doi.org/10.5281/zenodo.22666517>.
2. ———, *AlgebraicAnalysis v0.3.2*, Zenodo software release v0.3.2, 2026, DOI: <https://doi.org/10.5281/zenodo.22710109>.
3. ———, *Global Stafford*, Palomar Registry, <https://palomar-registry.org/entry?id=PALOMAR-2026-09-09-000001&version=2>, 2026, PALOMAR-2026-09-09-000001, version 2.
4. ———, *Global Stafford: Lean 4 formalization v1.0.6*, Zenodo software release v1.0.6, 2026, DOI: <https://doi.org/10.5281/zenodo.22669444>.
5. ———, *Stafford 3.8*, Palomar Registry, <https://palomar-registry.org/entry?id=PALOMAR-2026-09-05-000007&version=2>, 2026, PALOMAR-2026-09-05-000007, version 2.
6. ———, *Stafford's Conjecture 3.8 for Weyl algebras: Lean 4 formalization v1.2.2*, Zenodo software release v1.2.2, 2026, DOI: <https://doi.org/10.5281/zenodo.23079259>.
7. Gwyn Bellamy, *Module structure of Weyl algebras*, Journal of the London Mathematical Society **113** (2026), no. 1, e70373.
8. Jan-Erik Björk, *Rings of differential operators*, North-Holland Mathematical Library, vol. 21, North-Holland, Amsterdam, 1979.
9. Napoleon Caro and Daniel Levcovitz, *On a theorem of Stafford*, Cadernos de Matemática **11** (2010), 63–70, Preprint at <https://arxiv.org/abs/1005.4427>; journal PDF at the authors' university repository.
10. Napoleon Caro-Tuesta and Daniel Levcovitz, *Module structure of certain rings of differential operators*, Algebras and Representation Theory **23** (2020), 1637–1657.
11. Frédéric Chyzak, Alban Quadrat, and Daniel Robertz, *OreModules: a symbolic package for the study of multidimensional linear systems*, Applications of Time Delay Systems, Lecture Notes in Control and Information Sciences, vol. 352, Springer, 2007, pp. 233–264.
12. P. M. Cohn, *Algebra*, 2nd ed., vol. 1, John Wiley & Sons, Chichester, 1982.
13. Severino C Coutinho, *A primer of algebraic d-modules*, no. 33, Cambridge University Press, 1995.
14. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28 (Cham), Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
15. Ofer Gabber, *The integrability of the characteristic variety*, American Journal of Mathematics **103** (1981), no. 3, 445–468.
16. Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer, New York, 1977.
17. Andre Hillebrand and Wiland Schmale, *Towards an effective version of a theorem of Stafford*, Journal of Symbolic Computation **32** (2001), no. 6, 699–716.
18. Ryoshi Hotta, Kiyoshi Takeuchi, and Toshiyuki Tanisaki, *D-modules, perverse sheaves, and representation theory*, Progress in Mathematics, vol. 236, Birkhäuser, Boston, 2008.
19. LeanProver contributors, *Comparator: statement and proof comparison for Lean*, Project-pinned commit 575674928e239f5bc452aab72d1dd7b0f1326494. Source: <https://github.com/leanprover/comparator/tree/575674928e239f5bc452aab72d1dd7b0f1326494>.
20. Anton Leykin, *Algorithmic proofs of two theorems of Stafford*, Journal of Symbolic Computation **38** (2004), no. 6, 1535–1550.

21. Hideyuki Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics, vol. 8, Cambridge University Press, Cambridge, 1986, Translated from the Japanese by M. Reid.
22. Tadasi Nakayama, *A remark on finitely generated modules*, Nagoya Mathematical Journal **3** (1951), 139–140.
23. NanoDa: *an independent kernel checker for Lean*, Project-pinned commit 68d5ca9db226849b41a6fff59d796ff19d0a8840. Source: https://github.com/ammkr/n/nanoda_lib/tree/68d5ca9db226849b41a6fff59d796ff19d0a8840.
24. Øystein Ore, *Theory of non-commutative polynomials*, Annals of Mathematics **34** (1933), no. 3, 480–508.
25. Palomar Registry, *About the Palomar registry*, Official documentation, accessed 2026-09-30: <https://palomar-registry.org/about.html>.
26. Alban Quadrat and Daniel Robertz, *Computation of bases of free modules over the Weyl algebras*, Journal of Symbolic Computation **42** (2007), no. 11–12, 1113–1141.
27. ———, *A constructive study of the module structure of rings of partial differential operators*, Acta Applicandae Mathematicae **133** (2014), no. 1, 187–234.
28. Christian Schnell, *D-modules*, 2019, Lecture notes.
29. Jyoti Singh and Shiv Datt Kumar, *On the involutivity of the characteristic variety*, Communications in Algebra **42** (2014), no. 8, 3607–3618.
30. J. T. Stafford, *Module structure of Weyl algebras*, Journal of the London Mathematical Society. Second Series **18** (1978), no. 3, 429–442.
31. ———, *Generating modules efficiently: algebraic K-theory for noncommutative noetherian rings*, Journal of Algebra **69** (1981), no. 2, 312–346.
32. ———, *On the stable range of right noetherian rings*, Bulletin of the London Mathematical Society **13** (1981), no. 1, 39–41.
33. ———, *Generating modules efficiently over noncommutative rings*, Séminaire d’Algèbre Paul Dubreil et Marie-Paule Malliavin, Proceedings. Paris 1981 (34ème Année), Lecture Notes in Mathematics, vol. 924, Springer, Berlin, 1982, pp. 72–88.
34. ———, *Corrigendum*, Journal of Algebra **82** (1983), no. 1, 294–296, Corrigendum to Stafford’s 1981 article on generating modules efficiently.
35. ———, *Stably free, projective right ideals*, Compositio Mathematica **54** (1985), no. 1, 63–78, Available at https://www.numdam.org/item/CM_1985__54_1_63_0/.
36. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (New York), ACM, 2020, pp. 367–381.
37. The Stacks Project Authors, *Stacks project*, 2026.

APPENDIX A. AI-ASSISTED RESEARCH AND FORMALIZATION

Proof discovery, counterexample searches, review, and formalization was assisted by agentic AI tools. The work began with cyclic-extension computations in Macaulay2 and Maple. Attempts to construct universal cofactors directly produced special cases and reductions but eventually stalled. The decisive change was to study the canonical quotient $Q = A_n/(dA_n + x^N dA_n)$: Euler surjectivity and monicity constrained its characteristic support, while Gabber’s involutivity and asymptotic conormal geometry forced that support to be empty. This turned cofactor selection into a quotient-vanishing problem. Full Lean formalization was deferred until the paper argument was complete, although small formal checks continued throughout. In the final formal development, support avoidance over the coordinate hyperplane was proved by a localized filtered Koszul length contradiction.

The workflow separated proposed ideas from verified claims. One controller maintained the authoritative proof state, while parallel workers explored distinct mechanisms and returned evidence with an explicit scope. Independent reviewers checked fixed revisions for quantifiers, sidedness, circularity, and missing hypotheses. The project protocols were based on a `math-frontier` agent skill developed for this purpose: freeze the claim, identify its first unresolved dependency, and specify how a route could fail. Conceptual prompts requested a small number of distinct mechanisms; paper-only tasks sought complete arguments before implementation. A condensed representative prompt, adapted from the archived protocols, is:

Work on one precisely stated unresolved claim. Propose at most three distinct mechanisms, each with an intermediate lemma, a noncircular justification, and an adversarial test. Preserve the quantifiers and right-sided cofactors. For a paper task, supply a complete argument; for formalization, prove the required hypotheses without adding axioms or using `sorry`. Return evidence and the first unsupported bridge; do not promote a conditional result to a theorem.

Research and review sessions used GPT-5.5, GPT-5.6 Luna and Sol, GPT-6 Astra, and Claude Fable 5, Haiku 4.5 and Opus 5. The project protocols assigned routine implementation and replay to Luna and conceptual consultation to Sol; independent reports also record Opus reviews. Subsequent manuscript, proof-map, build and release work used Fable 5.1, Opus 5, Sonnet 5 and GPT-6 Luna. These records establish participation and task roles, without assigning a particular mathematical idea to one model.

The completed proof uses Lean 4 and Mathlib and pins AlgebraicAnalysis v0.3.0 [14, 36, 1]; the later library archive is v0.3.2 [2]. Formalization distinguished theorems proving required inputs from conditional theorems merely consuming them. Statement comparison and independent NanoDa and Lean kernel checks verified the submitted declarations [19, 23, 25]; their proofs use only `propext`, `Classical.choice`, and `Quot.sound`. The Stafford38 and Global Stafford headline proofs are registered separately in Palomar [5, 3]; their Lean developments are archived [6, 4].

APPENDIX B. FURTHER WORK

AI comment REF-01. Proposed change — author acceptance pending. The original appendix’s undefined labels `prop: symbol` and `eq: charinP` are visibly replaced below with the existing order-symbol and characteristic-containment labels.

~~AI also proved some related stuff regarding Right Ore localizations, the corresponding left-hand identity, and apparently the corresponding identity over other rings such as differential rings with laurent series coefficients; I am leaving that out for now.~~ The formalization also proves the following consequences, which are not otherwise used here: the identity in every right Ore localization of $A_n(k)$; the left-handed identity $1 = Rd + SdF$ via

the formal adjoint ; the identity for finite-order differential operators on localizations of $k[x_1, \dots, x_n]$, including principal opens, Laurent-polynomial coefficients and the fraction field ; and, independently of the geometric argument, an evolutionary version with arbitrary coefficients commuting with one Weyl pair .

AI comment SCOPE-01. Proposed change — author acceptance pending.

Review the scope before accepting this paragraph. The coefficients in the stated localization result are Laurent polynomials, not arbitrary Laurent series, and the operators have finite order. Retain further consequences only with their exact hypotheses and proof references; the declaration index is in the technical supplement.

B.1. Noncharacteristic H. An apparently nice ~~(though I do not understand its value)~~ side theorem that comes from "restriction to the coordinate hyper-plane" (after [Corollary 7.1](#)), is that $H = \{x_1 = 0\}$ is noncharacteristic:

The nonzero conormal covectors of H are the multiples $\lambda \frac{dx}{dx_1}$ (or in coordinates $(\lambda, 0, \dots, 0)$), with $\lambda \neq 0$. Since P is homogeneous and $P(1, 0, \dots, 0) = 1$, we have $P(\lambda, 0, \dots, 0) = \lambda^N \neq 0$, therefore $\lambda \frac{dx}{dx_1} \notin V(P)$. Since $\text{Char}(Q) \subseteq V(P)$ by [\(7.1\)](#), $\text{Char}(Q)$ therefore contains no nonzero conormal covector to H . More precisely, the projection

$$V(P) \cap T^*\mathbb{A}^n|_H \longrightarrow T^*H$$

is finite: on coordinate rings it adjoins ξ_1 subject to the monic equation $P(\xi_1, \dots, \xi_n) = 0$. Its restriction to $\text{Char}(Q)$ is therefore finite, which is the noncharacteristic condition [\[28, Definition 15.6\]](#).

Definition B.1 (Noncharacteristic, [\[28, Def. 15.6\]](#)). *Let $i: H \hookrightarrow X$ be a smooth closed subvariety and M a coherent $\mathcal{D}_X$ -module. Let $\varpi: T^*X|_H \rightarrow T^*H$ be the canonical surjection. Then H is noncharacteristic for M if the restriction $\varpi|_{\text{Char}(M) \cap T^*X|_H}: \text{Char}(M) \cap T^*X|_H \rightarrow T^*H$ is a finite morphism.*

Formal comparison: for $X = \mathbb{A}^n$, $H = \{x_1 = 0\}$ and a characteristic ideal J : the coordinate-ring map $k[T^*H] \rightarrow k[x, \xi]/(J + (x_1))$ is finite

Proposition B.2 (~~$H = \{x = 0\}$~~ $H = \{x_1 = 0\}$ is noncharacteristic for Q). *With P as in [\(4.2\)](#) and $H = \{x_1 = 0\}$, the projection $V(P) \cap T^*X|_H \rightarrow T^*H$ is finite. Hence H is noncharacteristic for Q . ~~Equivalently~~ Moreover, $\text{Char}(Q) \cap T_H^*X$ is contained in the zero section.*

Proof. Write $x' = (x_2, \dots, x_n)$, $\xi' = (\xi_2, \dots, \xi_n)$, so that $T^*X|_H = \text{Spec } k[x', \xi_1, \xi']$ and ϖ forgets ξ_1 . By [Proposition Lemma \ref{prop:symbol}4.5](#), P is monic of degree N in ξ_1 with coefficients in $k[\xi']$; therefore

$$k[x', \xi'] \longrightarrow k[x', \xi_1, \xi']/(P)$$

is a finite ring extension, generated by $1, \xi_1, \dots, \xi_1^{N-1}$. Thus $V(P) \cap T^*X|_H \rightarrow T^*H$ is finite, and by [\eqref{eq:charinP}\(7.1\)](#) so is its restriction to $\text{Char}(Q) \cap T^*X|_H$. For the last statement, the conormal space is $T_H^*X =$

$\{x_1 = 0, \xi' = 0\}$, on which $P(\xi_1, 0, \dots, 0) = \xi_1^N$; so $V(P) \cap T_H^*X = \{\xi = 0\}$ is the zero section. $\square$

Formal comparison: the zero-section statement holds without $x_1 = 0$: tangential covariables vanishing forces $\xi_1 = 0$

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